

维修手册_TC820LTYsi_SOP数据

维修手册知识库拆分数据 · 全字段 PDF 导出

源数据文件	维修手册_TC820LTYsi_SOP数据.jsonl
源文件路径	D:/QQ\维修手册_TC820LTYsi_SOP数据.jsonl
记录条数	35 条
字段并集	35 个字段（每条按分支实际字段渲染）
覆盖手册页码	106 – 97
文档来源	Southwestern Industries · TRAK TCsi with SINUMERIK ONE 维修手册
导出时间	2026-09-15 14:07:38

记录类别分布

记录类别	条数	占比
SOP 作业程序 (sop)	35	100.0%

知识类型分布

knowledge_type	条数
sop	35

说明：本 PDF 为该 JSONL 文件的逐条全字段渲染，未经任何筛选或删减；字段排序遵循 “标识与来源 / 型号与位置 / 分类与治理 / 业务明细 / 检索与原文” 的分组。空值以 “—” 表示，布尔值以 是/否 表示，数组按逐项换行展示。

#1 lty-sop-uncrate			SOP 作业程序
SOP编号	SOP-TC820LTysi-298	程序名称	开箱、吊运和放置
部件	machine installation	章节	TC820LTysi Model SOP
起始页	59	安全等级	critical
知识类型	sop	适用机型	TC820LTysi

摘要 | TC820LTysi 开箱、吊运和放置：按手册页码执行并以验收条件确认结果。

A • 标识与来源

记录ID	lty-sop-uncrate
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTysi
覆盖型号	TC820LTysi
型号范围	TC820LTysi
型号专属	是
章节编号	LTY-SOP
章节	TC820LTysi Model SOP
起始页	59
结束页	61

C • 分类与治理

记录类别	sop
知识类型	sop
部件	machine installation
安全等级	critical
可入库	是
需人工复核	是

D • 业务明细

前置条件	· 合格机器吊装人员执行 · 清理吊运路径并确认无人处于运动范围
安全警告	· 使用至少 15,000 lb 叉车 · 主轴箱侧较重，确认平衡 · 不得突然停止或提升过高

作业步骤	· 拆解木托盘并取出附件。 · 安装找平螺钉和外侧找平垫块。 · 将叉车货叉穿过底座，确认伸过远侧并检查平衡。 · 由第二人引导，缓慢放到找平垫块。 · 拆出冷却液箱、排屑机和包装支架。
验证方式	机器稳定、无倾覆和运输损伤。
验收标准	机器稳定、无倾覆和运输损伤。
SOP编号	SOP-TC820LTYsi-298
程序名称	开箱、吊运和放置
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	TC820LTYsi 开箱、吊运和放置：按手册页码执行并以验收条件确认结果。
关键词	· 15000 lb · 吊装 · 找平垫块
检索文本	TC820LTYsi; sop; sop; machine installation; 开箱、吊运和放置; TC820LTYsi 开箱、吊运和放置: 按手册页码执行并以验收条件确认结果。; 拆解木托盘并取出附件。; 安装找平螺钉和外侧找平垫块。; 将叉车货叉穿过底座，确认伸过远侧并检查平衡。; 由第二人引导，缓慢放到找平垫块。; 拆出冷却液箱、排屑机和包装支架。; 15000 lb; 吊装; 找平垫块; 3.14.2 TC820LTYsi Installation Checklist Installer – Use this checklist to assure a complete setup on the TC820LTYsi. Installation Checklist ITEMS TO BE COMPLETED BY QUALIFIED MACHINE RIGGER, IF NOT COMPLETED FOLLOW THE SUBSEQUENT STEPS: The Turning Center is packed on one wooden pallet during transportation: • • When arriving at the machine the rigger should have carefully disassembled the wooden pallet and removed all the loose parts from the machine. • • The leveling screws on the machine should be thread down by the rigger approximately 1.5” as measured from the bottom of the leveling screw to the bottom of the base casting. • • The rigger should have placed the machine on the 4 outside leveling pads. • The coolant tank is packed separately on the pallet; the rigger should first • remove any brackets holding the coolant tank in place, then remove the coolant tank from the pallet and set it aside. • The chip conveyor is packed separately on the pallet; the rigger should first • remove any brackets holding the chip conveyor in place, then remove the chip conveyor from the pallet and install the casters on swarf conveyer before placing it in the machine casting. Refer to section 3.3 Uncrating the TCsi, and 3.4 Lifting and Placing the TCsi in the Service Manual. The machine should be lifted and moved by a fork-lift truck • Use a fork-lift truck with a minimum loading capacity of 15,000 lbs to lift and move the machine. • Place the forks of the fork-lift under the machine. • Make sure the forks of the fork-lift protrude past the far edge of the machine base and ensure the machine is balanced before starting to lift and move it. • Pay special attention to the machine balance while lifting and moving, please note that the weight of the machine is greater on the headstock side. • Do not stop the lifting motions suddenly during the process and prevent sudden movement as the machine could become unbalanced. • Do not raise the machine too high as this may create a loss of stability and cause the fork-lift truck to overturn. • Have another person to help guide the way while lifting and moving the machine. Ensure nobody is in the way and the path is not blocked before moving the machine. • When placing the machine on leveling pads be careful when removing machine from the forklift truck, the machine is subject to tipping over if placed even a little off balance. Please contact TRAK customer service and a qualified machine rigger if you plan to lift and move the machine using an overhead crane. Only certified equipment should be

原文摘录

3.14.2 TC820LTYsi Installation Checklist Installer – Use this checklist to assure a complete setup on the TC820LTYsi. Installation Checklist ITEMS TO BE COMPLETED BY QUALIFIED MACHINE RIGGER, IF NOT COMPLETED FOLLOW THE SUBSEQUENT STEPS: The Turning Center is packed on one wooden pallet during transportation: • • When arriving at the machine the rigger should have carefully disassembled the wooden pallet and removed all the loose parts from the machine. • • The leveling screws on the machine should be thread down by the rigger approximately 1.5” as measured from the bottom of the leveling screw to the bottom of the base casting. • • The rigger should have placed the machine on the 4 outside leveling pads. • The coolant tank is packed separately on the pallet; the rigger should first • remove any brackets holding the coolant tank in place, then remove the coolant tank from the pallet and set it aside. • The chip conveyor is packed separately on the pallet; the rigger should first • remove any brackets holding the chip conveyor in place, then remove the chip conveyor from the pallet and install the casters on swarf conveyer before placing it in the machine casting. Refer to section 3.3 Uncrating the TCsi, and 3.4 Lifting and Placing the TCsi in the Service Manual. The machine should be lifted and moved by a fork-lift truck • Use a fork-lift truck with a minimum loading capacity of 15,000 lbs to lift and move the machine. • Place the forks of the fork-lift under the machine. • Make sure the forks of the fork-lift protrude past the far edge of the machine base and ensure the machine is balanced before starting to lift and move it. • Pay special attention to the machine balance while lifting and moving, please note that the weight of the machine is greater on the headstock side. • Do not stop the lifting motions suddenly during the process and prevent sudden movement as the machine could become unbalanced. • Do not raise the machine too high as this may create a loss of stability and cause the fork-lift truck to overturn. • Have another person to help guide the way while lifting and moving the machine. Ensure nobody is in the way and the path is not blocked before moving the machine. • When placing the machine on leveling pads be careful when removing machine from the forklift truck, the machine is subject to tipping over if placed even a little off balance. Please contact TRAK customer service and a qualified machine rigger if you plan to lift and move the machine using an overhead crane. Only certified equipment should be

#2 lty-sop-shipping-brackets			SOP 作业程序
SOP编号	SOP-TC820LTYsi-299	程序名称	拆除运输固定支架
部件	shipping brackets	章节	TC820LTYsi Model SOP
起始页	40	安全等级	critical
知识类型	sop	适用机型	TC820LTYsi

摘要 | TC820LTYsi 拆除运输固定支架：按手册页码执行并以验收条件确认结果。

A • 标识与来源

记录ID	lty-sop-shipping-brackets
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	TC820LTYsi

型号范围	TC820LTYsi
型号专属	是
章节编号	LTY-SOP
章节	TC820LTYsi Model SOP
起始页	40
结束页	42

C • 分类与治理

记录类别	sop
知识类型	sop
部件	shipping brackets
安全等级	critical
可入库	是
需人工复核	是

D • 业务明细

前置条件	机器稳定放置并可安全供电
安全警告	· 支架未拆前不得运行 · 不得手动移动自动尾座 · 支架和硬件不得丢弃
作业步骤	· 拆前门安全开关附近固定板。 · 从后方拆除 Z、X/Y 轴支架。 · 拆除尾座支架。 · 按手册步骤处理刀塔木块，不能直接手动取出。 · 将支架和硬件装袋放入工具箱。
验证方式	所有支架和木块已按步骤移除，支架已保存。
验收标准	所有支架和木块已按步骤移除，支架已保存。
SOP编号	SOP-TC820LTYsi-299
程序名称	拆除运输固定支架
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	TC820LTYsi 拆除运输固定支架：按手册页码执行并以验收条件确认结果。
关键词	· 运输支架 · 碰撞 · 尾座
检索文本	TC820LTYsi; sop; sop; shipping brackets; 拆除运输固定支架; TC820LTYsi 拆除运输固定支架：按手册页码执行并以验收条件确认结果。；拆前门安全开关附近固定板。；从后方拆除 Z、X/Y 轴支架。；拆除尾座支架。；按手册步骤处理刀塔木块，不能直接手动取出。；将支架和硬件装袋放入工具箱。；运输支架；碰撞；尾座； 3. Remove the bracket between the tailstock bed and tailstock base see figure 2.6.1c. Figure 2.6.1c Turret & Tailstock 4. Note the wooden block between the bed and the turret see figure 2.6.1c. DO NOT ATTEMPT TO REMOVE THE SUPPORT AT THIS TIME. Power needs to be supplied to move the X-Axis in the positive direction to remove the block. Make sure this is removed before using the machine! 3.6.2 TC820LTYsi Shipping Brackets On the TC820LTYsi machines there are shipping brackets that hold the axes in place during shipping. These brackets must be removed before operating the machine. Do not discard the shipping brackets. The shipping brackets must be replaced if the machine is ever moved.

原文摘录

3. Remove the bracket between the tailstock bed and tailstock base see figure 2.6.1c. Figure 2.6.1c Turret & Tailstock 4. Note the wooden block between the bed and the turret see figure 2.6.1c. DO NOT ATTEMPT TO REMOVE THE SUPPORT AT THIS TIME. Power needs to be supplied to move the X-Axis in the positive direction to remove the block. Make sure this is removed before using the machine! 3.6.2 TC820LTYsi Shipping Brackets On the TC820LTYsi machines there are shipping brackets that hold the axes in place during shipping. These brackets must be removed before operating the machine. Do not discard the shipping brackets. The shipping brackets must be replaced if the machine is ever moved.

#3 lty-sop-electrical			SOP 作业程序
SOP编号	SOP-TC820LTYsi-300	程序名称	电气接线和接地检查
部件	electrical system	章节	TC820LTYsi Model SOP
起始页	42	安全等级	critical
知识类型	sop	适用机型	TC820LTYsi

摘要 | TC820LTYsi 电气接线和接地检查：按手册页码执行并以验收条件确认结果。

A • 标识与来源

记录ID	lty-sop-electrical
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	TC820LTYsi
型号范围	TC820LTYsi
型号专属	是
章节编号	LTY-SOP
章节	TC820LTYsi Model SOP
起始页	42
结束页	60

C • 分类与治理

记录类别	sop
知识类型	sop
部件	electrical system
安全等级	critical
可入库	是

需人工复核	是
D • 业务明细	
前置条件	· 合格电工执行 · 电源已隔离且配置与铭牌一致
安全警告	· 三相 480 V; 432–528 V 可接受 · 现场核验 456–504 V · 约 8 ft 铜接地棒和 6 AWG 接地线
作业步骤	· 确认 480 V 或变压器 208 V 配置。 · 完成接地、进线密封和应力释放。 · 按图纸核对进线和断路器。 · 逐相测量并记录 456–504 V。 · 用相序表确认顺时针相序。
验证方式	电压、相序、接地和进线防护符合要求。
验收标准	电压、相序、接地和进线防护符合要求。
SOP编号	SOP-TC820LTYsi-300
程序名称	电气接线和接地检查
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	TC820LTYsi 电气接线和接地检查：按手册页码执行并以验收条件确认结果。
关键词	· 480V · 接地 · 相序 · 456-504V
检索文本	TC820LTYsi; sop; sop; electrical system; 电气接线和接地检查; TC820LTYsi 电气接线和接地检查：按手册页码执行并以验收条件确认结果。；确认 480 V 或变压器 208 V 配置。；完成接地、进线密封和应力释放。；按图纸核对进线和断路器。；逐相测量并记录 456–504 V。；用相序表确认顺时针相序。；480V; 接地; 相序; 456-504V; 3. Remove the bracket between the tailstock bed and tailstock base see figure 2.6.1c. Figure 2.6.1c Turret & Tailstock 4. Note the wooden block between the bed and the turret see figure 2.6.1c. DO NOT ATTEMPT TO REMOVE THE SUPPORT AT THIS TIME. Power needs to be supplied to move the X-Axis in the positive direction to remove the block. Make sure this is removed before using the machine! 3.7 Electrical Connection The TCsi machines come either with 208VAC input or 480VAC input. A 208VAC to 480VAC transformer is required for a 208VAC input. A 480VAC input does not require the transformer option. DANGER!!! Only a qualified electrician should wire the electricity for your machine installation. 3.7.1 Main Power to the Machine Connections The TCsi machine should be installed under the correct electrical environments. Before connecting the power wires, make sure the voltage is the same for both the machine and the plant power. Ensure the electrical cables have the same or better power rating as prescribed. Do not connect any power cable that might cause an abrupt voltage drop or generate signal noises on the power panel of the machine. Seal the electrical cabinet from moisture at the incoming power cable opening.

原文摘录

3. Remove the bracket between the tailstock bed and tailstock base see figure 2.6.1c. Figure 2.6.1c Turret & Tailstock 4. Note the wooden block between the bed and the turret see figure 2.6.1c. DO NOT ATTEMPT TO REMOVE THE SUPPORT AT THIS TIME. Power needs to be supplied to move the X-Axis in the positive direction to remove the block. Make sure this is removed before using the machine! 3.7 Electrical Connection The TCsi machines come either with 208VAC input or 480VAC input. A 208VAC to 480VAC transformer is required for a 208VAC input. A 480VAC input does not require the transformer option. DANGER!!! Only a qualified electrician should wire the electricity for your machine installation. 3.7.1 Main Power to the Machine Connections The TCsi machine should be installed under the correct electrical environments. Before connecting the power wires, make sure the voltage is the same for both the machine and the plant power. Ensure the electrical cables have the same or better power rating as prescribed. Do not connect any power cable that might cause an abrupt voltage drop or generate signal noises on the power panel of the machine. Seal the electrical cabinet from moisture at the incoming power cable opening.

#4 lty-sop-leveling			SOP 作业程序
SOP编号	SOP-TC820LTYsi-301	程序名称	机床找平和扭曲检查
部件	machine alignment	章节	TC820LTYsi Model SOP
起始页	45	安全等级	high
知识类型	sop	适用机型	TC820LTYsi

摘要 | TC820LTYsi 机床找平和扭曲检查：按手册页码执行并以验收条件确认结果。

A • 标识与来源

记录ID	lty-sop-leveling
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	TC820LTYsi
型号范围	TC820LTYsi
型号专属	是
章节编号	LTY-SOP
章节	TC820LTYsi Model SOP
起始页	45
结束页	47

C • 分类与治理

记录类别	sop
知识类型	sop

部件	machine alignment
安全等级	high
可入库	是
需人工复核	是

D • 业务明细

前置条件	· 运输支架已拆除 · 找平垫块稳定 · 使用校准水平仪
安全警告	· 水平 ≤ 0.0006 in/12 in · X/Z 扭曲 < 0.0006 in/12 in
作业步骤	· 从 Z 轴箱式导轨两侧测量。 · 将刀塔置于中心并逐点调整找平螺钉。 · 沿全 Z 行程检查 X/Z 扭曲。 · 锁紧螺母且不改变螺钉高度。
验证方式	多位置重复测量后仍在阈值内。
验收标准	多位置重复测量后仍在阈值内。
SOP编号	SOP-TC820LTYsi-301
程序名称	机床找平和扭曲检查
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	TC820LTYsi 机床找平和扭曲检查：按手册页码执行并以验收条件确认结果。
关键词	· 找平 · 扭曲 · 0.0006 in
检索文本	TC820LTYsi; sop; sop; machine alignment; 机床找平和扭曲检查; TC820LTYsi 机床找平和扭曲检查：按手册页码执行并以验收条件确认结果。；从 Z 轴箱式导轨两侧测量。；将刀塔置于中心并逐点调整找平螺钉。；沿全 Z 行程检查 X/Z 扭曲。；锁紧螺母且不改变螺钉高度。；找平；扭曲；0.0006 in；208 T BLACK WHITE L1 INCOMING S FAN FAN L2 L3 POWER R 480V L1 L2 L3 POWER TO MACHINE Figure 2.7.4 Transformer Wiring Configuration 220 Volt Options 208 T L1 L2 INCOMING S L3 480V POWER R BLACK WHITE POWER TO FAN FAN MACHINE L1 L2 L3 Figure 2.7.5 Transformer Wiring Configuration 240 Volt Options 3.8 Machine Leveling 3.8.1 Leveling Procedure The leveling of the TCsi machine consists of adjusting leveling screws (F1 – F4 and R1 – R4), shown in figure 2.8.1a, such that the machine is level using either analog or digital instruments. If the machine is not leveled properly, it may become twisted. Even a slight amount of twist will move centers out of alignment, and result in inaccurate work.
原文摘录	208 T BLACK WHITE L1 INCOMING S FAN FAN L2 L3 POWER R 480V L1 L2 L3 POWER TO MACHINE Figure 2.7.4 Transformer Wiring Configuration 220 Volt Options 208 T L1 L2 INCOMING S L3 480V POWER R BLACK WHITE POWER TO FAN FAN MACHINE L1 L2 L3 Figure 2.7.5 Transformer Wiring Configuration 240 Volt Options 3.8 Machine Leveling 3.8.1 Leveling Procedure The leveling of the TCsi machine consists of adjusting leveling screws (F1 – F4 and R1 – R4), shown in figure 2.8.1a, such that the machine is level using either analog or digital instruments. If the machine is not leveled properly, it may become twisted. Even a slight amount of twist will move centers out of alignment, and result in inaccurate work.

#5 lty-sop-coolant			SOP 作业程序
SOP编号	SOP-TC820LTYsi-302	程序名称	冷却液箱、冷却泵和冲屑泵安装
部件	coolant system	章节	TC820LTYsi Model SOP
起始页	47	安全等级	high
知识类型	sop	适用机型	TC820LTYsi

摘要 | TC820LTYsi 冷却液箱、冷却泵和冲屑泵安装：按手册页码执行并以验收条件确认结果。

A • 标识与来源

记录ID	lty-sop-coolant
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	TC820LTYsi
型号范围	TC820LTYsi
型号专属	是
章节编号	LTY-SOP
章节	TC820LTYsi Model SOP
起始页	47
结束页	48

C • 分类与治理

记录类别	sop
知识类型	sop
部件	coolant system
安全等级	high
可入库	是
需人工复核	是

D • 业务明细

前置条件	· 机床已找平 · 泵和接线处于断电状态
安全警告	· TC820LTYsi 使用 50 gallon 冷却液箱 · 普通泵接 U3、冲屑泵接 U4 · 旋向错误或泄漏不得运行

作业步骤	· 将冷却液箱滑入排屑机下方/右侧。 · 普通泵接 U3， chip wash pump 接 U4。 · 接软管并检查密封。 · 从 MCP 启动两泵，核对标签旋向。 · 确认冷却流经刀塔、冲屑流经 hose 且无泄漏。
验证方式	U3/U4 接线、旋向、流量和泄漏检查通过。
验收标准	U3/U4 接线、旋向、流量和泄漏检查通过。
SOP编号	SOP-TC820LTYsi-302
程序名称	冷却液箱、冷却泵和冲屑泵安装
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	TC820LTYsi 冷却液箱、冷却泵和冲屑泵安装：按手册页码执行并以验收条件确认结果。
关键词	· 50 gallon · U3 · U4 · chip wash
检索文本	TC820LTYsi; sop; sop; coolant system; 冷却液箱、冷却泵和冲屑泵安装; TC820LTYsi 冷却液箱、冷却泵和冲屑泵安装：按手册页码执行并以验收条件确认结果。; 将冷却液箱滑入排屑机下方/右侧。; 普通泵接 U3， chip wash pump 接 U4。; 接软管并检查密封。; 从 MCP 启动两泵，核对标签旋向。; 确认冷却流经刀塔、冲屑流经 hose 且无泄漏。; 50 gallon; U3; U4; chip wash; • Once the machine is sufficiently leveled lower screws F2, R2, F3 and R3 onto their leveling pads by hand and use a wrench and tighten only by 10°. Do not tighten any further as you will change the level of the machine. • Verify that the machine is still level by checking for twist at different points along the Z-Axis Box Ways. • Tighten the locking nuts for all leveling screws making sure that you do not raise or lower any of the leveling screws. • Verify that the machine is still level by checking for twist at different points along the Z-Axis Box Ways. Further Checks • To further check for twist in the machine a leveling plate can be attached to the turret with two calibrated bubble type levels or a multi axis digital level placed level on the plate, shown in figure 2.8.1.c. • Starting at the spindle of the machine, move the turret along the Z-Axis, stopping at 4 positions and allowing the bubble to settle. The change in level must be within 0.002" /10" . Figure 2.8.1c Leveling Plate 3.9 Placing the Coolant System The TC820si machine is fitted with a 33-gallon coolant tank and pump that delivers coolant through the turret at 30psi. The TC820LTYsi machine is fitted with a 50-gallon coolant tank and pump that delivers coolant through the turret at 70psi. Once the machine is leveled the coolant tank can be placed. For the TC820si machine, with the pump removed starting from the front of the machine, push the coolant tank through the center of the machine’ s base casting. The front face of the coolant tank should sit flush with the sheet metal guarding. For the TC820LTYsi machine, the coolant tank can slide in the right-hand side of the machine under the chip conveyor with the pumps located at the back of the machine.

原文摘录

• Once the machine is sufficiently leveled lower screws F2, R2, F3 and R3 onto their leveling pads by hand and use a wrench and tighten only by 10°. Do not tighten any further as you will change the level of the machine. • Verify that the machine is still level by checking for twist at different points along the Z-Axis Box Ways. • Tighten the locking nuts for all leveling screws making sure that you do not raise or lower any of the leveling screws. • Verify that the machine is still level by checking for twist at different points along the Z-Axis Box Ways. Further Checks • To further check for twist in the machine a leveling plate can be attached to the turret with two calibrated bubble type levels or a multi axis digital level placed level on the plate, shown in figure 2.8.1.c. • Starting at the spindle of the machine, move the turret along the Z-Axis, stopping at 4 positions and allowing the bubble to settle. The change in level must be within 0.002" /10" . Figure 2.8.1c Leveling Plate 3.9 Placing the Coolant System The TC820si machine is fitted with a 33-gallon coolant tank and pump that delivers coolant through the turret at 30psi. The TC820LTYsi machine is fitted with a 50-gallon coolant tank and pump that delivers coolant through the turret at 70psi. Once the machine is leveled the coolant tank can be placed. For the TC820si machine, with the pump removed starting from the front of the machine, push the coolant tank through the center of the machine’ s base casting. The front face of the coolant tank should sit flush with the sheet metal guarding. For the TC820LTYsi machine, the coolant tank can slide in the right-hand side of the machine under the chip conveyor with the pumps located at the back of the machine.

#6 lty-sop-conveyor			SOP 作业程序
SOP编号	SOP-TC820LTYsi-303	程序名称	排屑机安装和测试
部件	chip conveyor	章节	TC820LTYsi Model SOP
起始页	49	安全等级	high
知识类型	sop	适用机型	TC820LTYsi

摘要 | TC820LTYsi 排屑机安装和测试：按手册页码执行并以验收条件确认结果。

A • 标识与来源

记录ID	lty-sop-conveyor
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	TC820LTYsi
型号范围	TC820LTYsi
型号专属	是
章节编号	LTY-SOP
章节	TC820LTYsi Model SOP
起始页	49
结束页	49

C • 分类与治理

记录类别	sop
知识类型	sop
部件	chip conveyor
安全等级	high
可入库	是
需人工复核	是

D • 业务明细

前置条件	· 脚轮已安装 · 排屑机位于铸件内且运动区无人
安全警告	· 电缆接 U6 · 正反方向正确 · 反转按住 MCP 按键
作业步骤	· 安装脚轮并放置排屑机。 · 机电电缆接 U6并做应力释放。 · 用 MCP 测正转和反转。 · 安装盖板，使冷却液流入排屑机。
验证方式	正反转、回流和电缆布置正常。
验收标准	正反转、回流和电缆布置正常。
SOP编号	SOP-TC820LTYsi-303
程序名称	排屑机安装和测试
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	TC820LTYsi 排屑机安装和测试：按手册页码执行并以验收条件确认结果。
关键词	· U6 · 排屑机 · 正转 · 反转
检索文本	TC820LTYsi; sop; sop; chip conveyor; 排屑机安装和测试; TC820LTYsi 排屑机安装和测试：按手册页码执行并以验收条件确认结果。；安装脚轮并放置排屑机。；机电电缆接 U6并做应力释放。；用 MCP 测正转和反转。；安装盖板，使冷却液流入排屑机。；U6；排屑机；正转；反转；3.10 Setting Up Chip Conveyor The chip conveyor will be delivered installed in the machine envelope, make sure to install the wheels before placing the machine on the ground, see figure 2.10a. The chip conveyor is used to evacuate chips during machining. Connect the cable from the chip conveyor motor to U6 on the electrical cabinet side facing the coolant pump see figure 2.10b. To test the chip conveyor, use the forward and reverse buttons on the MCP and confirm the directions are correct. Figure 2.10a Chip Conveyor TCsi Figure 2.10b Chip Conveyor Connection TC820si Chip Conveyor Connection TC820LTYsi
原文摘录	3.10 Setting Up Chip Conveyor The chip conveyor will be delivered installed in the machine envelope, make sure to install the wheels before placing the machine on the ground, see figure 2.10a. The chip conveyor is used to evacuate chips during machining. Connect the cable from the chip conveyor motor to U6 on the electrical cabinet side facing the coolant pump see figure 2.10b. To test the chip conveyor, use the forward and reverse buttons on the MCP and confirm the directions are correct. Figure 2.10a Chip Conveyor TCsi Figure 2.10b Chip Conveyor Connection TC820si Chip Conveyor Connection TC820LTYsi

#7 lty-sop-footswitches			SOP 作业程序
SOP编号	SOP-TC820LTYsi-304	程序名称	卡盘和尾座套筒脚踏开关
部件	foot switches	章节	TC820LTYsi Model SOP
起始页	50	安全等级	high
知识类型	sop	适用机型	TC820LTYsi

摘要 | TC820LTYsi 卡盘和尾座套筒脚踏开关：按手册页码执行并以验收条件确认结果。

A • 标识与来源

记录ID	lty-sop-footswitches
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	TC820LTYsi
型号范围	TC820LTYsi
型号专属	是
章节编号	LTY-SOP
章节	TC820LTYsi Model SOP
起始页	50
结束页	50

C • 分类与治理

记录类别	sop
知识类型	sop
部件	foot switches
安全等级	high
可入库	是
需人工复核	是

D • 业务明细

前置条件	· 脚踏开关未接电 · 操作区域清洁
安全警告	· 测试卡盘时人员远离卡爪 · 电缆远离地面和运动件

作业步骤	· 将两个脚踏开关放在机器前方。 · 沿左侧布线并连接电柜下方连接器。 · 打开门测试卡盘夹紧/松开。 · 测试尾座套筒前进/后退。 · 核对卡盘 255 psi、限压 425 psi、套筒 70/140 psi。
验证方式	脚踏开关方向正确，压力符合安装要求。
验收标准	脚踏开关方向正确，压力符合安装要求。
SOP编号	SOP-TC820LTYsi-304
程序名称	卡盘和尾座套筒脚踏开关
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	TC820LTYsi 卡盘和尾座套筒脚踏开关：按手册页码执行并以验收条件确认结果。
关键词	· 脚踏开关 · 255 psi · 70 psi · 尾座套筒
检索文本	TC820LTYsi; sop; sop; foot switches; 卡盘和尾座套筒脚踏开关; TC820LTYsi 卡盘和尾座套筒脚踏开关: 按手册页码执行并以验收条件确认结果。; 将两个脚踏开关放在机器前方。; 沿左侧布线并连接电柜下方连接器。; 打开门测试卡盘夹紧/松开。; 测试尾座套筒前进/后退。; 核对卡盘 255 psi、限压 425 psi、套筒 70/140 psi。; 脚踏开关; 255 psi; 70 psi; 尾座套筒; 3.11 Setting Up Foot Switches 3.11.1 Setting Up Chuck Foot Switch The chuck foot switch will be delivered in a box along with the TCsi machine. Place the foot pedal in a convenient position for the operator in the front of the machine see figure 2.11a. Route the cable under the left side of the machine to the connector under the electrical cabinet close to the rear door. The connector only attaches one way, and the location is labeled inside the cabinet see figure 2.11b, inset photo installed cable under cabinet. To test its operation, with the door open press the foot pedal and the chuck jaws should actuate. Figure 2.11.1a Chuck Foot Switch Figure 2.11.1b Switch Connection 3.11.2 Setting Up Quill Foot Switch On the TC820LTYsi machines the quill foot switch will be delivered in a box along with the machine. Place the foot pedal in a convenient position for the operator in the front of the machine see figure 2.11.2a. Route the cable under the left side of the machine to the connector under the electrical cabinet close to the rear door. The connector only attaches one way, and the location is labeled inside the cabinet see figure 2.11.2b, inset photo installed cable under cabinet. To test its operation, press the foot pedal forward and the tailstock quill should move forward and press the foot pedal back and the tailstock quill should move back towards its original position. Figure 2.11.2a Quill Foot Switch Figure 2.11.2b Switch Connection
原文摘录	3.11 Setting Up Foot Switches 3.11.1 Setting Up Chuck Foot Switch The chuck foot switch will be delivered in a box along with the TCsi machine. Place the foot pedal in a convenient position for the operator in the front of the machine see figure 2.11a. Route the cable under the left side of the machine to the connector under the electrical cabinet close to the rear door. The connector only attaches one way, and the location is labeled inside the cabinet see figure 2.11b, inset photo installed cable under cabinet. To test its operation, with the door open press the foot pedal and the chuck jaws should actuate. Figure 2.11.1a Chuck Foot Switch Figure 2.11.1b Switch Connection 3.11.2 Setting Up Quill Foot Switch On the TC820LTYsi machines the quill foot switch will be delivered in a box along with the machine. Place the foot pedal in a convenient position for the operator in the front of the machine see figure 2.11.2a. Route the cable under the left side of the machine to the connector under the electrical cabinet close to the rear door. The connector only attaches one way, and the location is labeled inside the cabinet see figure 2.11.2b, inset photo installed cable under cabinet. To test its operation, press the foot pedal forward and the tailstock quill should move forward and press the foot pedal back and the tailstock quill should move back towards its original position. Figure 2.11.2a Quill Foot Switch Figure 2.11.2b Switch Connection

#8 lty-sop-hydraulic-lube			SOP 作业程序
SOP编号	SOP-TC820LTYsi-305	程序名称	液压和润滑系统调试
部件	hydraulic and lubrication	章节	TC820LTYsi Model SOP
起始页	61	安全等级	high
知识类型	sop	适用机型	TC820LTYsi

摘要 | TC820LTYsi 液压和润滑系统调试：按手册页码执行并以验收条件确认结果。

A • 标识与来源

记录ID	lty-sop-hydraulic-lube
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	TC820LTYsi
型号范围	TC820LTYsi
型号专属	是
章节编号	LTY-SOP
章节	TC820LTYsi Model SOP
起始页	61
结束页	62

C • 分类与治理

记录类别	sop
知识类型	sop
部件	hydraulic and lubrication
安全等级	high
可入库	是
需人工复核	是

D • 业务明细

前置条件	· 完成接线、找平和支架拆除 · 液压/润滑油箱可安全检查
安全警告	· 压力异常立即停机 · 保持远离运动件和高压油路

作业步骤	· 释放急停并按 Reset 启动液压泵。 · 检查液压油位、ISO VG32 和泄漏。 · 按 MCP 润滑键 5 次运行润滑泵。 · 确认润滑压力 70 psi。 · 点动 X/Y/Z，确认导轨和丝杠见油。
验证方式	液压稳定、无泄漏、润滑到达导轨和丝杠。
验收标准	液压稳定、无泄漏、润滑到达导轨和丝杠。
SOP编号	SOP-TC820LTYsi-305
程序名称	液压和润滑系统调试
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	TC820LTYsi 液压和润滑系统调试：按手册页码执行并以验收条件确认结果。
关键词	· 700 psi · 70 psi · ISO VG32 · ISO VG68
检索文本	TC820LTYsi; sop; sop; hydraulic and lubrication; 液压和润滑系统调试; TC820LTYsi 液压和润滑系统调试：按手册页码执行并以验收条件确认结果。; 释放急停并按 Reset 启动液压泵。; 检查液压油位、ISO VG32 和泄漏。; 按 MCP 润滑键 5 次运行润滑泵。; 确认润滑压力 70 psi。; 点动 X/Y/Z，确认导轨和丝杠见油。; 700 psi; 70 psi; ISO VG32; ISO VG68; Installation Checklist (8) Cast Iron Leveling Pads (May already be placed under the machine) (2) Sets of 3 pendant keys 8 Verify 2 electrical cabinet keys are zip tied to the electrical cabinet door lock on the • inside of the electrical cabinet. Remove covers and shipping brackets: • On the TC820LTYsi machines there are shipping brackets that hold the axes in place during shipping, these brackets must be removed before operating the machine, see section 2.6.2 of the Service Manual. • Remove the sheet metal cover on rear of the machine exposing the Z- 9 axis box ways and the X- • axis and the Y-axis casting. • Remove the Z-axis shipping bracket. • Remove the X/Y-axis shipping bracket. • Remove the sheet metal cover on the tailstock side of the machine exposing the rear of the tailstock box ways. • Remove the tailstock shipping bracket. • DO NOT MOVE THE TAILSTOCK BY HAND, THIS WILL CAUSE IT TO LOSE POSITION AND MAY CAUSE A CRASH. • Remove the front door shipping bracket. • Place the shipping brackets and hardware in a bag and store in the machine toolbox. 10 Clean the machine if needed and remove any remaining protective grease. • Turn on the power to the machine. To start machine after boot up: • Release all E-Stop buttons. 11 • Press and hold reset button, at this point you should hear the hydraulic pump • and system start (the pump should rotate CCW per the label on the pump). • Turn Feed Start on to allow axis motion. • Turn Spindle Start on to allow spindle motion. Check the level of the machine from both sides of the Z-axis box ways. Center the turret 12 as required before leveling. The machine should be level to within 0.0006"/12" . Refer to • section 3.8 of the Service Manual. 13 Check the X and Z-axis twist from the turret across the entire Z-axis travel. The machine • twist should be < 0.0006" /12" . 14 Confirm the MCP panel has the key plugged in to change the access level. Key is • shipped in toolbox. Check the Hydraulic Fluid Tank level and fill with ISO VG 32 oil if needed. • Inspect all hydraulic lines, fittings and components to make sure there are no • leaks. 15 Run the lube pump by pressing the lube button on MCP 5 times. •

原文摘录

Installation Checklist (8) Cast Iron Leveling Pads (May already be placed under the machine) (2) Sets of 3 pendant keys 8 Verify 2 electrical cabinet keys are zip tied to the electrical cabinet door lock on the • inside of the electrical cabinet. Remove covers and shipping brackets: • On the TC820LTYsi machines there are shipping brackets that hold the axes in place during shipping, these brackets must be removed before operating the machine, see section 2.6.2 of the Service Manual. • Remove the sheet metal cover on rear of the machine exposing the Z- 9 axis box ways and the X- • axis and the Y-axis casting. • Remove the Z-axis shipping bracket. • Remove the X/Y-axis shipping bracket. • Remove the sheet metal cover on the tailstock side of the machine exposing the rear of the tailstock box ways. • Remove the tailstock shipping bracket. • DO NOT MOVE THE TAILSTOCK BY HAND, THIS WILL CAUSE IT TO LOSE POSITION AND MAY CAUSE A CRASH. • Remove the front door shipping bracket. • Place the shipping brackets and hardware in a bag and store in the machine toolbox. 10 Clean the machine if needed and remove any remaining protective grease. • Turn on the power to the machine. To start machine after boot up: • Release all E-Stop buttons. 11 • Press and hold reset button, at this point you should hear the hydraulic pump • and system start (the pump should rotate CCW per the label on the pump). • Turn Feed Start on to allow axis motion. • Turn Spindle Start on to allow spindle motion. Check the level of the machine from both sides of the Z-axis box ways. Center the turret 12 as required before leveling. The machine should be level to within 0.0006"/12" . Refer to • section 3.8 of the Service Manual. 13 Check the X and Z-axis twist from the turret across the entire Z-axis travel. The machine • twist should be < 0.0006" /12" . 14 Confirm the MCP panel has the key plugged in to change the access level. Key is • shipped in toolbox. Check the Hydraulic Fluid Tank level and fill with ISO VG 32 oil if needed. • Inspect all hydraulic lines, fittings and components to make sure there are no • leaks. 15 Run the lube pump by pressing the lube button on MCP 5 times. •

#9 lty-sop-acceptance			SOP 作业程序
SOP编号	SOP-TC820LTYsi-306	程序名称	开机验收、TAILSTOCK_TEST 和 TURRET_TEST
部件	machine acceptance	章节	TC820LTYsi Model SOP
起始页	64	安全等级	critical
知识类型	sop	适用机型	TC820LTYsi

摘要 | TC820LTYsi 开机验收、TAILSTOCK_TEST 和 TURRET_TEST：按手册页码执行并以验收条件确认结果。

A • 标识与来源

记录ID	lty-sop-acceptance
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	TC820LTYsi
型号范围	TC820LTYsi

型号专属	是
章节编号	LTY-SOP
章节	TC820LTYsi Model SOP
起始页	64
结束页	66

C • 分类与治理

记录类别	sop
知识类型	sop
部件	machine acceptance
安全等级	critical
可入库	是
需人工复核	是

D • 业务明细

前置条件	· 安装和安全检查完成 · 人员远离运动区
安全警告	· 急停、门互锁、主轴制动失败不得生产 · 持续观察噪声、泄漏和状态灯
作业步骤	· 测试 MPG、X/Y/Z 方向、软件限位、急停、门互锁、工作灯和主轴制动。 · 测试尾座销、停车开关、工件开关和限位。 · 运行 TAILSTOCK_TEST 15 分钟。 · 运行 TURRET_TEST，验证正反分度和刀具表工位。 · 测试探头、主轴 100–4000 RPM 和动力刀具 100–4500 RPM。
验证方式	安全功能、尾座、刀塔、探头、主轴和动力刀具均有记录。
验收标准	安全功能、尾座、刀塔、探头、主轴和动力刀具均有记录。
SOP编号	SOP-TC820LTYsi-306
程序名称	开机验收、TAILSTOCK_TEST 和 TURRET_TEST
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	TC820LTYsi 开机验收、TAILSTOCK_TEST 和 TURRET_TEST：按手册页码执行并以验收条件确认结果。
关键词	· TAILSTOCK_TEST · TURRET_TEST · 100-4000 RPM · 4500 RPM

检索文本	TC820LTYsi; sop; sop; machine acceptance; 开机验收、TAILSTOCK_TEST 和 TURRET_TEST; TC820LTYsi 开机验收、TAILSTOCK_TEST 和 TURRET_TEST: 按手册页码执行并以验收条件确认结果。; 测试 MPG、X/Y/Z 方向、软件限位、急停、门互锁、工作灯和主轴制动。; 测试尾座销、停车开关、工件开关和限位。; 运行 TAILSTOCK_TEST 15 分钟。; 运行 TURRET_TEST, 验证正反分度和刀具表工位。; 测试探头、主轴 100–4000 RPM 和动力刀具 100–4500 RPM。; TAILSTOCK_TEST; TURRET_TEST; 100-4000 RPM; 4500 RPM; Installation Checklist • Inspect all coolant lines, fittings and components to make sure there are no leaks. Check spindle motor cooling: • Remove the sheet metal cover to the spindle motor fan • Make sure the air is blowing towards the motor. 21 • Check electrical cabinet cooling: • Open the electrical cabinet door. 22 • Make sure the air is blowing into the heat exchange on both sides of the door. • Make sure the cabinet circulation fan is blowing towards the components in the electrical cabinet. Check MPG functions correctly: • Move all axis at all speed settings and confirm each axis moves in the correct 22 direction. • • Confirm Deadman switch on MPG is working to move the axis when the door is open. Test machine MCP motion and limits: • Set the MPG to off. • Test each axis by using the Axis + and - button with and without the RAPID 23 button on the MCP. • • Verify the machine stops on the software limits before hitting the hard stop. Adjust parameters 36100 (-) and 36110 (+) on the axis in question if you do hit a hard stop. Check to make sure that the E-Stop button is functioning correctly: 24 • Turn spindle on and jog an axis and press the E-stop button during this • operation and verify the spindle and axis stops. Test door interlock: 25 • Close the door and make sure the control recognizes the door as being closed. • • Once you close the door, you should not be able to open it until you press the door open button on the MCP.
原文摘录	Installation Checklist • Inspect all coolant lines, fittings and components to make sure there are no leaks. Check spindle motor cooling: • Remove the sheet metal cover to the spindle motor fan • Make sure the air is blowing towards the motor. 21 • Check electrical cabinet cooling: • Open the electrical cabinet door. 22 • Make sure the air is blowing into the heat exchange on both sides of the door. • Make sure the cabinet circulation fan is blowing towards the components in the electrical cabinet. Check MPG functions correctly: • Move all axis at all speed settings and confirm each axis moves in the correct 22 direction. • • Confirm Deadman switch on MPG is working to move the axis when the door is open. Test machine MCP motion and limits: • Set the MPG to off. • Test each axis by using the Axis + and - button with and without the RAPID 23 button on the MCP. • • Verify the machine stops on the software limits before hitting the hard stop. Adjust parameters 36100 (-) and 36110 (+) on the axis in question if you do hit a hard stop. Check to make sure that the E-Stop button is functioning correctly: 24 • Turn spindle on and jog an axis and press the E-stop button during this • operation and verify the spindle and axis stops. Test door interlock: 25 • Close the door and make sure the control recognizes the door as being closed. • • Once you close the door, you should not be able to open it until you press the door open button on the MCP.

#10 lty-sop-tailstock-reference		SOP 作业程序	
SOP编号	SOP-TC820LTYsi-307	程序名称	自动尾座参考位置恢复
部件	automatic tailstock	章节	TC820LTYsi Model SOP
起始页	117	安全等级	critical
知识类型	sop	适用机型	TC820LTYsi

摘要 | TC820LTYsi 自动尾座参考位置恢复: 按手册页码执行并以验收条件确认结果。

A • 标识与来源

记录ID	lty-sop-tailstock-reference
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf

制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	TC820LTYsi
型号范围	TC820LTYsi
型号专属	是
章节编号	LTY-SOP
章节	TC820LTYsi Model SOP
起始页	117
结束页	117

C • 分类与治理

记录类别	sop
知识类型	sop
部件	automatic tailstock
安全等级	critical
可入库	是
需人工复核	是

D • 业务明细

前置条件	· 确认自动尾座丢失位置 · 尾座和 Z 轴无干涉
安全警告	· 禁止手动推动尾座 · 尾座销与 pickup hole 对齐后操作
作业步骤	· 移动 Z 轴使尾座销对齐 pickup hole。 · 用 MCP 松开尾座。 · 用 MCP 使尾座销啮合开关。 · 退回尾座销并夹紧尾座。
验证方式	控制器重新学习尾座相对 Z 轴位置。
验收标准	控制器重新学习尾座相对 Z 轴位置。
SOP编号	SOP-TC820LTYsi-307
程序名称	自动尾座参考位置恢复
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	TC820LTYsi 自动尾座参考位置恢复：按手册页码执行并以验收条件确认结果。
关键词	· pickup hole · 参考位置 · 尾座销

检索文本

TC820LTYsi; sop; sop; automatic tailstock; 自动尾座参考位置恢复; TC820LTYsi
自动尾座参考位置恢复: 按手册页码执行并以验收条件确认结果。; 移动 Z 轴使尾座销对齐 pickup
hole。; 用 MCP 松开尾座。; 用 MCP 使尾座销啮合开关。; 退回尾座销并夹紧尾座。; pickup
hole; 参考位置; 尾座销; 7.3.1 Automatic Tailstock Has Lost its Position It is important that the
TC820LTYsi tailstock is not moved by hand. If the tailstock is moved by hand the control will lose its
position and will not be able to locate it for automatic tailstock commands. If the control loses the
position of the automatic tailstock: • Move the Z-axis to line the tailstock pin up with the pickup
hole on the Z-Axis. There are markers on the Z-Axis and the Tailstock body that show when this is
lined up. • Unclamp the tailstock using the button on the MCP. • Engage the tailstock pin using the
button on the MCP. The pin must engage the switch in the pickup hole to learn the new position. •
Disengage the tailstock pin using the button on the MCP. • Clamp the tailstock using the button on
the MCP. The machine should now have learned the position of the tailstock relative to the z-axis
machine coordinate system. 7.3.2 Automatic Tailstock in JOG Mode To correctly position the
tailstock manually in jog mode: • Move the Z-axis to line the tailstock pin up with either the park
hole or pick up hole (depending on where they want to move the tailstock to). • Unclamp the
tailstock using the button on MCP button or M code. • Engage the tailstock pin using the MCP
button or M code. • Move the Z-axis to position the tailstock in the location desired. • Disengage
the tailstock pin using the MCP button or M code. • Clamp the tailstock using the MCP button or M
code. o The tailstock location can be found from the z-axis position in the machine coordinate
system or from the custom screen (shown below) in the programing environment that can be used
for automatically positioning the tailstock in a part program. • Set the quill pressure. • Position the
quill in the desired position. • Open the tailstock cover to move the quill sensor to match the
required position. o The user would do this to set the tailstock position before running an
automatic cycle. From this they will have the quill position set and a tailstock location they can use
in their part program. • Close the tailstock cover to make the machine ready for machining. • The
tailstock can be manually put on the park switch following the first 6 steps of the method outlined
above and moving it back onto the park switch. • The tailstock can automatically be moved to the
park switch using a m-code. 7.3.3 Automatic Tailstock in Auto Mode To correctly position the
tailstock manually in auto mode: If the user wants to automatically move the tailstock in a part
program, they must do it by using the custom screen. For this the tailstock position must be known
by the machine and the user must know where they want to position the tailstock relative to the
Z-axis machine coordinate system. To use the automatic tailstock cycle in a part program the user
must: • Go to the Tailstock custom screen in the part programming environment. o The tailstock
custom screen can be found by pressing the tailstock button on the second page of the horizontal
soft keys. • ZP: New Tailstock Position

原文摘录

7.3.1 Automatic Tailstock Has Lost its Position It is important that the TC820LTYsi tailstock is not moved by hand. If the tailstock is moved by hand the control will lose its position and will not be able to locate it for automatic tailstock commands. If the control loses the position of the automatic tailstock: • Move the Z-axis to line the tailstock pin up with the pickup hole on the Z-Axis. There are markers on the Z-Axis and the Tailstock body that show when this is lined up. • Unclamp the tailstock using the button on the MCP. • Engage the tailstock pin using the button on the MCP. The pin must engage the switch in the pickup hole to learn the new position. • Disengage the tailstock pin using the button on the MCP. • Clamp the tailstock using the button on the MCP. The machine should now have learned the position of the tailstock relative to the z-axis machine coordinate system.

7.3.2 Automatic Tailstock in JOG Mode To correctly position the tailstock manually in jog mode: • Move the Z-axis to line the tailstock pin up with either the park hole or pick up hole (depending on where they want to move the tailstock to). • Unclamp the tailstock using the button on MCP button or M code. • Engage the tailstock pin using the MCP button or M code. • Move the Z-axis to position the tailstock in the location desired. • Disengage the tailstock pin using the MCP button or M code. • Clamp the tailstock using the MCP button or M code. o The tailstock location can be found from the z-axis position in the machine coordinate system or from the custom screen (shown below) in the programing environment that can be used for automatically positioning the tailstock in a part program. • Set the quill pressure. • Position the quill in the desired position. • Open the tailstock cover to move the quill sensor to match the required position. o The user would do this to set the tailstock position before running an automatic cycle. From this they will have the quill position set and a tailstock location they can use in their part program. • Close the tailstock cover to make the machine ready for machining. • The tailstock can be manually put on the park switch following the first 6 steps of the method outlined above and moving it back onto the park switch. • The tailstock can automatically be moved to the park switch using a m-code.

7.3.3 Automatic Tailstock in Auto Mode To correctly position the tailstock manually in auto mode: If the user wants to automatically move the tailstock in a part program, they must do it by using the custom screen. For this the tailstock position must be known by the machine and the user must know where they want to position the tailstock relative to the Z-axis machine coordinate system. To use the automatic tailstock cycle in a part program the user must: • Go to the Tailstock custom screen in the part programming environment. o The tailstock custom screen can be found by pressing the tailstock button on the second page of the horizontal soft keys. • ZP: New Tailstock Position

#11 lty-sop-tailstock-jog		SOP 作业程序	
SOP编号	SOP-TC820LTYsi-308	程序名称	自动尾座 JOG 模式定位
部件	automatic tailstock	章节	TC820LTYsi Model SOP
起始页	117	安全等级	high
知识类型	sop	适用机型	TC820LTYsi

摘要 | TC820LTYsi 自动尾座 JOG 模式定位：按手册页码执行并以验收条件确认结果。

A • 标识与来源

记录ID	lty-sop-tailstock-jog
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	TC820LTYsi
型号范围	TC820LTYsi
型号专属	是
章节编号	LTY-SOP
章节	TC820LTYsi Model SOP
起始页	117
结束页	117

C • 分类与治理

记录类别	sop
知识类型	sop
部件	automatic tailstock
安全等级	high
可入库	是
需人工复核	是

D • 业务明细

前置条件	· 主轴停止且路径清空 · 确定 park hole 或 pickup hole
安全警告	· 不允许手动推移 · 啮合/退销前不得移动 Z 轴
作业步骤	· 对齐目标孔位。 · 松开尾座并啮合尾座销。 · 移动 Z 轴到目标位置。 · 退销并夹紧。 · 设置套筒压力和位置，必要时调整传感器。
验证方式	尾座位置、park/part switch 与实际状态一致。
验收标准	尾座位置、park/part switch 与实际状态一致。
SOP编号	SOP-TC820LTYsi-308
程序名称	自动尾座 JOG 模式定位
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	TC820LTYsi 自动尾座 JOG 模式定位：按手册页码执行并以验收条件确认结果。
关键词	· JOG · park hole · pickup hole

检索文本

TC820LTYsi; sop; sop; automatic tailstock; 自动尾座 JOG 模式定位; TC820LTYsi 自动尾座 JOG 模式定位: 按手册页码执行并以验收条件确认结果。; 对齐目标孔位。; 松开尾座并啮合尾座销。; 移动 Z 轴到目标位置。; 退销并夹紧。; 设置套筒压力和位置, 必要时调整传感器。; JOG; park hole; pickup hole; 7.3.1 Automatic Tailstock Has Lost its Position It is important that the TC820LTYsi tailstock is not moved by hand. If the tailstock is moved by hand the control will lose its position and will not be able to locate it for automatic tailstock commands. If the control loses the position of the automatic tailstock: • Move the Z-axis to line the tailstock pin up with the pickup hole on the Z-Axis. There are markers on the Z-Axis and the Tailstock body that show when this is lined up. • Unclamp the tailstock using the button on the MCP. • Engage the tailstock pin using the button on the MCP. The pin must engage the switch in the pickup hole to learn the new position. • Disengage the tailstock pin using the button on the MCP. • Clamp the tailstock using the button on the MCP. The machine should now have learned the position of the tailstock relative to the z-axis machine coordinate system. 7.3.2 Automatic Tailstock in JOG Mode To correctly position the tailstock manually in jog mode: • Move the Z-axis to line the tailstock pin up with either the park hole or pick up hole (depending on where they want to move the tailstock to). • Unclamp the tailstock using the button on MCP button or M code. • Engage the tailstock pin using the MCP button or M code. • Move the Z-axis to position the tailstock in the location desired. • Disengage the tailstock pin using the MCP button or M code. • Clamp the tailstock using the MCP button or M code. o The tailstock location can be found from the z-axis position in the machine coordinate system or from the custom screen (shown below) in the programming environment that can be used for automatically positioning the tailstock in a part program. • Set the quill pressure. • Position the quill in the desired position. • Open the tailstock cover to move the quill sensor to match the required position. o The user would do this to set the tailstock position before running an automatic cycle. From this they will have the quill position set and a tailstock location they can use in their part program. • Close the tailstock cover to make the machine ready for machining. • The tailstock can be manually put on the park switch following the first 6 steps of the method outlined above and moving it back onto the park switch. • The tailstock can automatically be moved to the park switch using a m-code. 7.3.3 Automatic Tailstock in Auto Mode To correctly position the tailstock manually in auto mode: If the user wants to automatically move the tailstock in a part program, they must do it by using the custom screen. For this the tailstock position must be known by the machine and the user must know where they want to position the tailstock relative to the Z-axis machine coordinate system. To use the automatic tailstock cycle in a part program the user must: • Go to the Tailstock custom screen in the part programming environment. o The tailstock custom screen can be found by pressing the tailstock button on the second page of the horizontal soft keys. • ZP: New Tailstock Position

原文摘录

7.3.1 Automatic Tailstock Has Lost its Position It is important that the TC820LTYsi tailstock is not moved by hand. If the tailstock is moved by hand the control will lose its position and will not be able to locate it for automatic tailstock commands. If the control loses the position of the automatic tailstock: • Move the Z-axis to line the tailstock pin up with the pickup hole on the Z-Axis. There are markers on the Z-Axis and the Tailstock body that show when this is lined up. • Unclamp the tailstock using the button on the MCP. • Engage the tailstock pin using the button on the MCP. The pin must engage the switch in the pickup hole to learn the new position. • Disengage the tailstock pin using the button on the MCP. • Clamp the tailstock using the button on the MCP. The machine should now have learned the position of the tailstock relative to the z-axis machine coordinate system.

7.3.2 Automatic Tailstock in JOG Mode To correctly position the tailstock manually in jog mode: • Move the Z-axis to line the tailstock pin up with either the park hole or pick up hole (depending on where they want to move the tailstock to). • Unclamp the tailstock using the button on MCP button or M code. • Engage the tailstock pin using the MCP button or M code. • Move the Z-axis to position the tailstock in the location desired. • Disengage the tailstock pin using the MCP button or M code. • Clamp the tailstock using the MCP button or M code. o The tailstock location can be found from the z-axis position in the machine coordinate system or from the custom screen (shown below) in the programing environment that can be used for automatically positioning the tailstock in a part program. • Set the quill pressure. • Position the quill in the desired position. • Open the tailstock cover to move the quill sensor to match the required position. o The user would do this to set the tailstock position before running an automatic cycle. From this they will have the quill position set and a tailstock location they can use in their part program. • Close the tailstock cover to make the machine ready for machining. • The tailstock can be manually put on the park switch following the first 6 steps of the method outlined above and moving it back onto the park switch. • The tailstock can automatically be moved to the park switch using a m-code.

7.3.3 Automatic Tailstock in Auto Mode To correctly position the tailstock manually in auto mode: If the user wants to automatically move the tailstock in a part program, they must do it by using the custom screen. For this the tailstock position must be known by the machine and the user must know where they want to position the tailstock relative to the Z-axis machine coordinate system. To use the automatic tailstock cycle in a part program the user must: • Go to the Tailstock custom screen in the part programming environment. o The tailstock custom screen can be found by pressing the tailstock button on the second page of the horizontal soft keys. • ZP: New Tailstock Position

#12 lty-sop-tailstock-auto		SOP 作业程序	
SOP编号	SOP-TC820LTYsi-309	程序名称	自动尾座 AUTO 模式定位
部件	automatic tailstock	章节	TC820LTYsi Model SOP
起始页	117	安全等级	high
知识类型	sop	适用机型	TC820LTYsi

摘要 | TC820LTYsi 自动尾座 AUTO 模式定位：按手册页码执行并以验收条件确认结果。

A • 标识与来源

记录ID	lty-sop-tailstock-auto
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	TC820LTYsi
型号范围	TC820LTYsi
型号专属	是
章节编号	LTY-SOP
章节	TC820LTYsi Model SOP
起始页	117
结束页	118

C • 分类与治理

记录类别	sop
知识类型	sop
部件	automatic tailstock
安全等级	high
可入库	是
需人工复核	是

D • 业务明细

前置条件	· 控制器已知尾座参考位置 · 首件已确定 Z 位置和套筒位置
安全警告	· 验证 ZP/QL，避免进入工件或刀具路径 · 停车前确认空间和限位逻辑
作业步骤	· 进入 Tailstock custom screen。 · 输入 ZP 目标位置。 · 在 QL 选择 Forward 或 Back。 · 确认 ZPOS 当前值并选择 Accept。 · 需要停车时使用手册 M-Code。
验证方式	程序尾座定位与首件设置一致。
验收标准	程序尾座定位与首件设置一致。
SOP编号	SOP-TC820LTYsi-309
程序名称	自动尾座 AUTO 模式定位
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	TC820LTYsi 自动尾座 AUTO 模式定位：按手册页码执行并以验收条件确认结果。
关键词	· AUTO · ZP · QL · ZPOS

检索文本

TC820LTYsi; sop; sop; automatic tailstock; 自动尾座 AUTO 模式定位; TC820LTYsi 自动尾座 AUTO 模式定位: 按手册页码执行并以验收条件确认结果。; 进入 Tailstock custom screen。; 输入 ZP 目标位置。; 在 QL 选择 Forward 或 Back。; 确认 ZPOS 当前值并选择 Accept。; 需要停车时使用手册 M-Code。; AUTO; ZP; QL; ZPOS; 7.3.1 Automatic Tailstock Has Lost its Position It is important that the TC820LTYsi tailstock is not moved by hand. If the tailstock is moved by hand the control will lose its position and will not be able to locate it for automatic tailstock commands. If the control loses the position of the automatic tailstock: • Move the Z-axis to line the tailstock pin up with the pickup hole on the Z-Axis. There are markers on the Z-Axis and the Tailstock body that show when this is lined up. • Unclamp the tailstock using the button on the MCP. • Engage the tailstock pin using the button on the MCP. The pin must engage the switch in the pickup hole to learn the new position. • Disengage the tailstock pin using the button on the MCP. • Clamp the tailstock using the button on the MCP. The machine should now have learned the position of the tailstock relative to the z-axis machine coordinate system. 7.3.2 Automatic Tailstock in JOG Mode To correctly position the tailstock manually in jog mode: • Move the Z-axis to line the tailstock pin up with either the park hole or pick up hole (depending on where they want to move the tailstock to). • Unclamp the tailstock using the button on MCP button or M code. • Engage the tailstock pin using the MCP button or M code. • Move the Z-axis to position the tailstock in the location desired. • Disengage the tailstock pin using the MCP button or M code. • Clamp the tailstock using the MCP button or M code. o The tailstock location can be found from the z-axis position in the machine coordinate system or from the custom screen (shown below) in the programing environment that can be used for automatically positioning the tailstock in a part program. • Set the quill pressure. • Position the quill in the desired position. • Open the tailstock cover to move the quill sensor to match the required position. o The user would do this to set the tailstock position before running an automatic cycle. From this they will have the quill position set and a tailstock location they can use in their part program. • Close the tailstock cover to make the machine ready for machining. • The tailstock can be manually put on the park switch following the first 6 steps of the method outlined above and moving it back onto the park switch. • The tailstock can automatically be moved to the park switch using a m-code. 7.3.3 Automatic Tailstock in Auto Mode To correctly position the tailstock manually in auto mode: If the user wants to automatically move the tailstock in a part program, they must do it by using the custom screen. For this the tailstock position must be known by the machine and the user must know where they want to position the tailstock relative to the Z-axis machine coordinate system. To use the automatic tailstock cycle in a part program the user must: • Go to the Tailstock custom screen in the part programming environment. o The tailstock custom screen can be found by pressing the tailstock button on the second page of the horizontal soft keys. • ZP: New Tailstock Position

原文摘录

7.3.1 Automatic Tailstock Has Lost its Position It is important that the TC820LTYsi tailstock is not moved by hand. If the tailstock is moved by hand the control will lose its position and will not be able to locate it for automatic tailstock commands. If the control loses the position of the automatic tailstock: • Move the Z-axis to line the tailstock pin up with the pickup hole on the Z-Axis. There are markers on the Z-Axis and the Tailstock body that show when this is lined up. • Unclamp the tailstock using the button on the MCP. • Engage the tailstock pin using the button on the MCP. The pin must engage the switch in the pickup hole to learn the new position. • Disengage the tailstock pin using the button on the MCP. • Clamp the tailstock using the button on the MCP. The machine should now have learned the position of the tailstock relative to the z-axis machine coordinate system.

7.3.2 Automatic Tailstock in JOG Mode To correctly position the tailstock manually in jog mode: • Move the Z-axis to line the tailstock pin up with either the park hole or pick up hole (depending on where they want to move the tailstock to). • Unclamp the tailstock using the button on MCP button or M code. • Engage the tailstock pin using the MCP button or M code. • Move the Z-axis to position the tailstock in the location desired. • Disengage the tailstock pin using the MCP button or M code. • Clamp the tailstock using the MCP button or M code. o The tailstock location can be found from the z-axis position in the machine coordinate system or from the custom screen (shown below) in the programing environment that can be used for automatically positioning the tailstock in a part program. • Set the quill pressure. • Position the quill in the desired position. • Open the tailstock cover to move the quill sensor to match the required position. o The user would do this to set the tailstock position before running an automatic cycle. From this they will have the quill position set and a tailstock location they can use in their part program. • Close the tailstock cover to make the machine ready for machining. • The tailstock can be manually put on the park switch following the first 6 steps of the method outlined above and moving it back onto the park switch. • The tailstock can automatically be moved to the park switch using a m-code.

7.3.3 Automatic Tailstock in Auto Mode To correctly position the tailstock manually in auto mode: If the user wants to automatically move the tailstock in a part program, they must do it by using the custom screen. For this the tailstock position must be known by the machine and the user must know where they want to position the tailstock relative to the Z-axis machine coordinate system. To use the automatic tailstock cycle in a part program the user must: • Go to the Tailstock custom screen in the part programming environment. o The tailstock custom screen can be found by pressing the tailstock button on the second page of the horizontal soft keys. • ZP: New Tailstock Position

#13 lty-sop-tailstock-sensor			SOP 作业程序
SOP编号	SOP-TC820LTYsi-310	程序名称	尾座套筒传感器定位
部件	tailstock sensors	章节	TC820LTYsi Model SOP
起始页	119	安全等级	high
知识类型	sop	适用机型	TC820LTYsi

摘要 | TC820LTYsi 尾座套筒传感器定位：按手册页码执行并以验收条件确认结果。

A • 标识与来源

记录ID	lty-sop-tailstock-sensor
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	TC820LTYsi
型号范围	TC820LTYsi
型号专属	是
章节编号	LTY-SOP
章节	TC820LTYsi Model SOP
起始页	119
结束页	120

C • 分类与治理

记录类别	sop
知识类型	sop
部件	tailstock sensors
安全等级	high
可入库	是
需人工复核	是

D • 业务明细

前置条件	设备停止且确认需要调整传感器
安全警告	· 清洁后拆罩，防止切屑进入 · 保护垫片和线缆
作业步骤	· 确认 700427 或相关位置消息。 · 清洁并拆下四个罩螺钉。 · 松开六角螺母滑动传感器座。 · 使指示灯在正确位置点亮并紧固。 · 整理线缆、清洁垫片并复装。
验证方式	前进/退回反馈与实际位置一致。
验收标准	前进/退回反馈与实际位置一致。
SOP编号	SOP-TC820LTYsi-310
程序名称	尾座套筒传感器定位
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	TC820LTYsi 尾座套筒传感器定位：按手册页码执行并以验收条件确认结果。
关键词	· 700427 · 接近开关 · 传感器

检索文本	TC820LTYsi; sop; sop; tailstock sensors; 尾座套筒传感器定位; TC820LTYsi 尾座套筒传感器定位：按手册页码执行并以验收条件确认结果。；确认 700427 或相关位置消息。；清洁并拆下四个罩螺钉。；松开六角螺母滑动传感器座。；使指示灯在正确位置点亮并紧固。；整理线缆、清洁垫片并复装。；700427; 接近开关; 传感器; 7.3.4 Extended Tail Stock Will Not Respond to Control Sensors inside the tailstock body can be used to determine the tailstock quill position and motion during operation. In order to set the tailstock quill, position the operator will have to adjust a sensor under the tailstock cover. See figure 6.3.2a showing the tailstock position indicator and the control measurement sensors. For the control to allow a cycle start command the tailstock quill must be sat on its backward sensor. The control will display the message, "700427 Tailstock Quill is no located on the Advanced or retracted Proximity Switch" . Figure 6.3.2a shows the details of the forward and backward sensors which need positioning. Figure 6.3.2a Tailstock Sensors In order to set the tailstock quill position sensors, first clean off the cover and surrounding areas to prevent dirt and debris from entering tailstock. In order to position the sensors with the tailstock quill, remove the four knurled screws that hold the cover in position see figure 6.3.2b. Be careful with the gasket, it needs to be cleaned and reused. Figure 6.3.2b Tailstock Sensor Access Cover
原文摘录	7.3.4 Extended Tail Stock Will Not Respond to Control Sensors inside the tailstock body can be used to determine the tailstock quill position and motion during operation. In order to set the tailstock quill, position the operator will have to adjust a sensor under the tailstock cover. See figure 6.3.2a showing the tailstock position indicator and the control measurement sensors. For the control to allow a cycle start command the tailstock quill must be sat on its backward sensor. The control will display the message, "700427 Tailstock Quill is no located on the Advanced or retracted Proximity Switch" . Figure 6.3.2a shows the details of the forward and backward sensors which need positioning. Figure 6.3.2a Tailstock Sensors In order to set the tailstock quill position sensors, first clean off the cover and surrounding areas to prevent dirt and debris from entering tailstock. In order to position the sensors with the tailstock quill, remove the four knurled screws that hold the cover in position see figure 6.3.2b. Be careful with the gasket, it needs to be cleaned and reused. Figure 6.3.2b Tailstock Sensor Access Cover

#14 lty-sop-tailstock-alignment		SOP 作业程序	
SOP编号	SOP-TC820LTYsi-311	程序名称	尾座中心线、Z 轴和 B 轴对中
部件	tailstock alignment	章节	TC820LTYsi Model SOP
起始页	115	安全等级	high
知识类型	sop	适用机型	TC820LTYsi

摘要 | TC820LTYsi 尾座中心线、Z 轴和 B 轴对中：按手册页码执行并以验收条件确认结果。

A • 标识与来源

记录ID	lty-sop-tailstock-alignment
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	TC820LTYsi

型号范围	TC820LTYsi
型号专属	是
章节编号	LTY-SOP
章节	TC820LTYsi Model SOP
起始页	115
结束页	116

C • 分类与治理

记录类别	sop
知识类型	sop
部件	tailstock alignment
安全等级	high
可入库	是
需人工复核	是

D • 业务明细

前置条件	· 碰撞或加工锥度时执行 · 测量面清洁
安全警告	· 调整前必须有精密测量设备 · 小量调整并复测
作业步骤	· 用千分表扫测尾座套筒孔，中心线误差 ≤ 0.002 in。 · 用 MT4 对中杆和磁性表座检查 Z 平行度。 · 顶部/侧面误差 < 0.001 in/4 in。 · B 轴移到另一端重复检查。 · 超差时用调整螺栓小量修正。
验证方式	中心线、Z 和 B 轴平行度达到阈值。
验收标准	中心线、Z 和 B 轴平行度达到阈值。
SOP编号	SOP-TC820LTYsi-311
程序名称	尾座中心线、Z 轴和 B 轴对中
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	TC820LTYsi 尾座中心线、Z 轴和 B 轴对中：按手册页码执行并以验收条件确认结果。
关键词	· MT4 · 0.002 in · 0.001 in/4 in

检索文本

TC820LTYsi; sop; sop; tailstock alignment; 尾座中心线、Z 轴和 B 轴对中; TC820LTYsi

尾座中心线、Z 轴和 B

轴对中: 按手册页码执行并以验收条件确认结果。; 用千分表扫描尾座套筒孔, 中心线误差 ≤ 0.002

in。; 用 MT4 对中杆和磁性表座检查 Z 平行度。; 顶部/侧面误差 < 0.001 in/4 in。; B

轴移到另一端重复检查。; 超差时用调整螺栓小量修正。; MT4; 0.002 in; 0.001 in/4 in; 7.2 Tailstock

Alignment 7.2.1 Tailstock to Spindle Centerline Alignment The machine should have a common

centerline between the spindle, turret pocket at the X0 position, and the tailstock. After a crash

occurs the spindle to tailstock centerline should be checked and adjusted accordingly. The

maximum permitted error is $0.002''$ (0.05mm). To Check the current spindle to turret pocket

centerline alignment: 1. Clamp an indicator in the chuck and move the tailstock so the indicator

feeler is measuring the pocket of the tailstock quill. 2. Make sure that the measuring surfaces are

clean. Any debris or dirt will affect the measurements. 3. Sweep the pocket of the tailstock quill and

measure the error in the pocket. The error must not exceed $0.002''$ (0.05mm). 4. If this value is

exceeded you will need to adjust your tailstock centerline. 7.2.2 Tailstock to Z-Axis Parallelism It is

important when using the tailstock, it is aligned to the Z-Axis motion on the machine. To check the

tailstock parallelism: 1. Install a MT4 taper alignment tool into the tailstock quill. 2. Attach a

magnetic base to the turret body. 3. Set up your indicator so that it is reading off the side of the

alignment bar. Be sure to have your indicator set up in a way to reduce any cosine errors present in

the reading. 4. Make sure that the measuring surfaces are clean. Any debris or dirt will affect the

measurements. 5. Move the turret in the Z direction so your indicator runs across the top of the

alignment bar. 6. Record the difference in position on your indicator from the start to the end of

travel along the alignment bar. 7. Repeat this process taking a reading from the top of the

alignment bar. 8. The error along the top and the side of the alignment bar should be less than

$0.001''/4''$ (0.025mm/100mm). 9. If the indicator shows an error of greater than this your tailstock

needs realigning to make it move parallel to the Z-Axis motion in the machine. NOTE: if you do not

have a MT4 alignment bar, place a precision ground bar between centers on the headstock and

tailstock measuring the error along the top and side of the precision ground bar, see figure 6.2.2.

The measured error across the span should be $< 0.001''/4''$. Figure 6.2.2 Tailstock Alignment

Between Centers

原文摘录

7.2 Tailstock Alignment 7.2.1 Tailstock to Spindle Centerline Alignment The machine should have a

common centerline between the spindle, turret pocket at the X0 position, and the tailstock. After a

crash occurs the spindle to tailstock centerline should be checked and adjusted accordingly. The

maximum permitted error is $0.002''$ (0.05mm). To Check the current spindle to turret pocket

centerline alignment: 1. Clamp an indicator in the chuck and move the tailstock so the indicator

feeler is measuring the pocket of the tailstock quill. 2. Make sure that the measuring surfaces are

clean. Any debris or dirt will affect the measurements. 3. Sweep the pocket of the tailstock quill and

measure the error in the pocket. The error must not exceed $0.002''$ (0.05mm). 4. If this value is

exceeded you will need to adjust your tailstock centerline. 7.2.2 Tailstock to Z-Axis Parallelism It is

important when using the tailstock, it is aligned to the Z-Axis motion on the machine. To check the

tailstock parallelism: 1. Install a MT4 taper alignment tool into the tailstock quill. 2. Attach a

magnetic base to the turret body. 3. Set up your indicator so that it is reading off the side of the

alignment bar. Be sure to have your indicator set up in a way to reduce any cosine errors present in

the reading. 4. Make sure that the measuring surfaces are clean. Any debris or dirt will affect the

measurements. 5. Move the turret in the Z direction so your indicator runs across the top of the

alignment bar. 6. Record the difference in position on your indicator from the start to the end of

travel along the alignment bar. 7. Repeat this process taking a reading from the top of the

alignment bar. 8. The error along the top and the side of the alignment bar should be less than

$0.001''/4''$ (0.025mm/100mm). 9. If the indicator shows an error of greater than this your tailstock

needs realigning to make it move parallel to the Z-Axis motion in the machine. NOTE: if you do not

have a MT4 alignment bar, place a precision ground bar between centers on the headstock and

tailstock measuring the error along the top and side of the precision ground bar, see figure 6.2.2.

The measured error across the span should be $< 0.001''/4''$. Figure 6.2.2 Tailstock Alignment

Between Centers

#15 lty-sop-turret-reference			SOP 作业程序
SOP编号	SOP-TC820LTYsi-312	程序名称	刀塔工位重新参考和同步
部件	turret	章节	TC820LTYsi Model SOP
起始页	106	安全等级	high
知识类型	sop	适用机型	TC820LTYsi

摘要 | TC820LTYsi 刀塔工位重新参考和同步：按手册页码执行并以验收条件确认结果。

A • 标识与来源

记录ID	lty-sop-turret-reference
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	TC820LTYsi
型号范围	TC820LTYsi
型号专属	是
章节编号	LTY-SOP
章节	TC820LTYsi Model SOP
起始页	106
结束页	107

C • 分类与治理

记录类别	sop
知识类型	sop
部件	turret
安全等级	high
可入库	是
需人工复核	是

D • 业务明细

前置条件	· 实际工位与刀具表不一致 · 刀塔周围无干涉
安全警告	· 进入 service mode 前确认主轴/尾座安全 · 退出前确认刀塔夹紧

作业步骤	· MCP 按 JOG 和 REF PT。 · 控制器进入 turret menu 执行 Turret Reference。 · Feed Start 快按三次进入 service mode。 · 查看 IP/OP Clamp 状态，必要时用 T REF/T PLUS/T Minus。 · 退出前确认夹紧且工位一致。
验证方式	刀塔工位与刀具表绿色工位一致。
验收标准	刀塔工位与刀具表绿色工位一致。
SOP编号	SOP-TC820LTYsi-312
程序名称	刀塔工位重新参考和同步
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	TC820LTYsi 刀塔工位重新参考和同步：按手册页码执行并以验收条件确认结果。
关键词	· T REF · REF PT · 刀塔参考
检索文本	TC820LTYsi; sop; sop; turret; 刀塔工位重新参考和同步; TC820LTYsi 刀塔工位重新参考和同步：按手册页码执行并以验收条件确认结果。； MCP 按 JOG 和 REF PT。； 控制器进入 turret menu 执行 Turret Reference。； Feed Start 快按三次进入 service mode。； 查看 IP/OP Clamp 状态，必要时用 T REF/T PLUS/T Minus。； 退出前确认夹紧且工位一致。； T REF； REF PT； 刀塔参考； 6.0 Sauter 12 Station Turret 6.1 12 Station Hydraulic Turret The TCsi series is fitted with a 12-station turret from Sauter. The TC820si uses and Bolt-on Turret face and the TC820LTYsi is fitted with a BMT45 Turret face with a live tooling drive. The BMT45 turret for the TC820LTY is available in a half index option providing 24 indexing positions. The turret is driven by an AC servo motor and utilizes a hydraulic clamping mechanism and proximity switch. As the tool exchange command is issued, turret will be rotated to the selected tool position according to NC or manual commands. When the selected tool traces the target location, the AC servo motor will stop immediately. The hydraulic system will then lock the tool magazine to prevent the tool magazine from further traveling. The proximity switch is used for checking the magazine’ s clamping. The rotation direction could be either cw or ccw. The tool is selected based on the shortest path to minimize the tool selection time. 6.2 Troubleshooting the 12 Station Turret 6.2.1 12 Station Turret Out of Sync with Control Display Occasionally the turret position will become out-of-synch with the control display. Typically, this is when the green highlighted entry of the Tool List as shown in figure 5.2.1a shows a different number than the tool position number engraved into the turret body at the current cutting position. Figure 5.2.1a Tool List Showing Tool Position To re-sync the turret with the tool table, on the MCP press “JOG” and “REF PT” . On the control press 1 - “>” arrow to pull out hidden window, 2 – turret menu, then 3 – Turret Reference as shown on figure 5.2.1b.
原文摘录	6.0 Sauter 12 Station Turret 6.1 12 Station Hydraulic Turret The TCsi series is fitted with a 12-station turret from Sauter. The TC820si uses and Bolt-on Turret face and the TC820LTYsi is fitted with a BMT45 Turret face with a live tooling drive. The BMT45 turret for the TC820LTY is available in a half index option providing 24 indexing positions. The turret is driven by an AC servo motor and utilizes a hydraulic clamping mechanism and proximity switch. As the tool exchange command is issued, turret will be rotated to the selected tool position according to NC or manual commands. When the selected tool traces the target location, the AC servo motor will stop immediately. The hydraulic system will then lock the tool magazine to prevent the tool magazine from further traveling. The proximity switch is used for checking the magazine’ s clamping. The rotation direction could be either cw or ccw. The tool is selected based on the shortest path to minimize the tool selection time. 6.2 Troubleshooting the 12 Station Turret 6.2.1 12 Station Turret Out of Sync with Control Display Occasionally the turret position will become out-of-synch with the control display. Typically, this is when the green highlighted entry of the Tool List as shown in figure 5.2.1a shows a different number than the tool position number engraved into the turret body at the current cutting position. Figure 5.2.1a Tool List Showing Tool Position To re-sync the turret with the tool table, on the MCP press “JOG” and “REF PT” . On the control press 1 - “>” arrow to pull out hidden window, 2 – turret menu, then 3 – Turret Reference as shown on figure 5.2.1b.

#16 lty-sop-turret-parallelism			SOP 作业程序
SOP编号	SOP-TC820LTYsi-313	程序名称	刀塔 X/Z 轴平行度检查
部件	turret alignment	章节	TC820LTYsi Model SOP
起始页	108	安全等级	high
知识类型	sop	适用机型	TC820LTYsi

摘要 | TC820LTYsi 刀塔 X/Z 轴平行度检查：按手册页码执行并以验收条件确认结果。

A • 标识与来源

记录ID	lty-sop-turret-parallelism
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	TC820LTYsi
型号范围	TC820LTYsi
型号专属	是
章节编号	LTY-SOP
章节	TC820LTYsi Model SOP
起始页	108
结束页	110

C • 分类与治理

记录类别	sop
知识类型	sop
部件	turret alignment
安全等级	high
可入库	是
需人工复核	是

D • 业务明细

前置条件	· 碰撞后或精度异常时执行 · 测量面清洁
安全警告	· 调整两个螺钉会同时影响 X/Z · 超差由精密测量人员处理

作业步骤	· 磁性表座安装于尾座导轨。 · 沿 X 轴约 4 in 扫测刀塔面。 · 沿 Z 轴约 4 in 扫测周边面。 · 两轴误差均需 < 0.0004 in/4 in。 · 按手册调整后交替复测。
验证方式	X/Z 平行度均达标。
验收标准	X/Z 平行度均达标。
SOP编号	SOP-TC820LTYsi-313
程序名称	刀塔 X/Z 轴平行度检查
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	TC820LTYsi 刀塔 X/Z 轴平行度检查：按手册页码执行并以验收条件确认结果。
关键词	· 0.0004 in · X轴 · Z轴
检索文本	TC820LTYsi; sop; sop; turret alignment; 刀塔 X/Z 轴平行度检查; TC820LTYsi 刀塔 X/Z 轴平行度检查：按手册页码执行并以验收条件确认结果。; 磁性表座安装于尾座导轨。; 沿 X 轴约 4 in 扫测刀塔面。; 沿 Z 轴约 4 in 扫测周边面。; 两轴误差均需 < 0.0004 in/4 in。; 按手册调整后交替复测。; 0.0004 in; X轴; Z轴; 6.2.2 Turret Parallelism to Axis Motion It is essential that the turret is set up and aligned parallel to the axis motion of the machine in order to ensure proper accuracy. The turret parallelism to the axis motion should be checked after a crash has occurred, see figure 5.2.2. Figure 5.2.2 Ready to Measure 6.2.2.1 Turret to X-Axis Parallelism To check the turret parallelism to the X-Axis motion: 1. Attach a magnetic base to a tailstock way, see figure 5.2.2. 2. Set up your indicator so that it is reading off the turret’ s face where a turning tool will be mounted, see figure 5.2.2.1. Be sure to have your indicator set up in a way to reduce any cosine errors present in the reading. 3. Make sure that the measuring surfaces on the turret are clean. Any debris or dirt on the turret will affect the measurements. 4. Move the turret in the X direction so your indicator runs across its face. You should be able to achieve about 4 inches of travel. 5. Record the difference in position on your indicator from the start to the end of travel along the turret face. 6. The error should be less than 0.0004” /4” (0.010mm/100mm). 7. If the indicator shows an error of greater than this your turret needs realigning to make it move parallel to the X-Axis motion in the machine.
原文摘录	6.2.2 Turret Parallelism to Axis Motion It is essential that the turret is set up and aligned parallel to the axis motion of the machine in order to ensure proper accuracy. The turret parallelism to the axis motion should be checked after a crash has occurred, see figure 5.2.2. Figure 5.2.2 Ready to Measure 6.2.2.1 Turret to X-Axis Parallelism To check the turret parallelism to the X-Axis motion: 1. Attach a magnetic base to a tailstock way, see figure 5.2.2. 2. Set up your indicator so that it is reading off the turret’ s face where a turning tool will be mounted, see figure 5.2.2.1. Be sure to have your indicator set up in a way to reduce any cosine errors present in the reading. 3. Make sure that the measuring surfaces on the turret are clean. Any debris or dirt on the turret will affect the measurements. 4. Move the turret in the X direction so your indicator runs across its face. You should be able to achieve about 4 inches of travel. 5. Record the difference in position on your indicator from the start to the end of travel along the turret face. 6. The error should be less than 0.0004” /4” (0.010mm/100mm). 7. If the indicator shows an error of greater than this your turret needs realigning to make it move parallel to the X-Axis motion in the machine.

#17 lty-sop-centerline			SOP 作业程序
SOP编号	SOP-TC820LTYsi-314	程序名称	主轴与刀塔中心线同心度
部件	spindle and turret alignment	章节	TC820LTYsi Model SOP
起始页	111	安全等级	high
知识类型	sop	适用机型	TC820LTYsi

摘要 | TC820LTYsi 主轴与刀塔中心线同心度：按手册页码执行并以验收条件确认结果。

A • 标识与来源

记录ID	lty-sop-centerline
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	TC820LTYsi
型号范围	TC820LTYsi
型号专属	是
章节编号	LTY-SOP
章节	TC820LTYsi Model SOP
起始页	111
结束页	111

C • 分类与治理

记录类别	sop
知识类型	sop
部件	spindle and turret alignment
安全等级	high
可入库	是
需人工复核	是

D • 业务明细

前置条件	· 碰撞后或加工异常时执行 · 刀塔和测量面清洁
安全警告	· 卡盘夹表确认夹持可靠 · 调整由精密测量人员执行

作业步骤	· 安装镗刀杆夹持座并移至 X0。 · 卡盘夹表，扫测夹持座孔。 · 按 ±X/±Y 位置记录总误差。 · 确认总误差 ≤ 0.002 in，超差则调整并复测。
验证方式	刀塔孔相对主轴中心线总误差达标。
验收标准	刀塔孔相对主轴中心线总误差达标。
SOP编号	SOP-TC820LTYsi-314
程序名称	主轴与刀塔中心线同心度
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	TC820LTYsi 主轴与刀塔中心线同心度：按手册页码执行并以验收条件确认结果。
关键词	· 0.002 in · 同心度 · 碰撞
检索文本	TC820LTYsi; sop; sop; spindle and turret alignment; 主轴与刀塔中心线同心度; TC820LTYsi 主轴与刀塔中心线同心度：按手册页码执行并以验收条件确认结果。；安装镗刀杆夹持座并移至 X0。；卡盘夹表，扫测夹持座孔。；按 ±X/±Y 位置记录总误差。；确认总误差 ≤ 0.002 in，超差则调整并复测。；0.002 in；同心度；碰撞；The machine should have a common centerline between the spindle, turret pocket at the X0 position, and the tailstock. After a crash occurs the spindle to turret pocket centerline should be checked and adjusted accordingly. The maximum permitted error is 0.002" (0.05mm). To Check the current spindle to turret pocket centerline alignment: • Install a boring bar holder and move the turret so it is at X0,. • Make sure that the mating surfaces of the turret and tooling are clean. Any debris between the turret and the boring bar holder will affect the measurements. • Clamp an indicator in the chuck and move the turret so the indicator feeler is measuring the pocket of the boring bar holder. • Set your indicator zero at position 1, +X. Refer to figure 5.2.2.4a to show the positions of ±X and ±Y. • Sweep the pocket of the boring bar holder and measure the error in the pocket. Refer to figure 5.2.2.4b to show the actual tooling used. • The total error must not exceed 0.002" (0.05mm). • If the total error exceeds this value you will need to adjust your turret centerline position. Figure 5.2.2.4a Spindle to Turret Centerline Figure 5.2.2.4b Tooling for Spindle to Turret Centerline 6.2.2.5 TC820LTY Driven Tooling Drive Alignment.
原文摘录	The machine should have a common centerline between the spindle, turret pocket at the X0 position, and the tailstock. After a crash occurs the spindle to turret pocket centerline should be checked and adjusted accordingly. The maximum permitted error is 0.002" (0.05mm). To Check the current spindle to turret pocket centerline alignment: • Install a boring bar holder and move the turret so it is at X0,. • Make sure that the mating surfaces of the turret and tooling are clean. Any debris between the turret and the boring bar holder will affect the measurements. • Clamp an indicator in the chuck and move the turret so the indicator feeler is measuring the pocket of the boring bar holder. • Set your indicator zero at position 1, +X. Refer to figure 5.2.2.4a to show the positions of ±X and ±Y. • Sweep the pocket of the boring bar holder and measure the error in the pocket. Refer to figure 5.2.2.4b to show the actual tooling used. • The total error must not exceed 0.002" (0.05mm). • If the total error exceeds this value you will need to adjust your turret centerline position. Figure 5.2.2.4a Spindle to Turret Centerline Figure 5.2.2.4b Tooling for Spindle to Turret Centerline 6.2.2.5 TC820LTY Driven Tooling Drive Alignment.

#18 lty-sop-driven-tool			SOP 作业程序
SOP编号	SOP-TC820LTYsi-315	程序名称	动力刀具驱动对中
部件	live tooling	章节	TC820LTYsi Model SOP
起始页	112	安全等级	critical
知识类型	sop	适用机型	TC820LTYsi

摘要 | TC820LTYsi 动力刀具驱动对中：按手册页码执行并以验收条件确认结果。

A • 标识与来源

记录ID	lty-sop-driven-tool
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	TC820LTYsi
型号范围	TC820LTYsi
型号专属	是
章节编号	LTY-SOP
章节	TC820LTYsi Model SOP
起始页	112
结束页	112

C • 分类与治理

记录类别	sop
知识类型	sop
部件	live tooling
安全等级	critical
可入库	是
需人工复核	是

D • 业务明细

前置条件	· 动力刀具不能正确啮合 · 移除动力刀具和前盖
安全警告	· 保持运动区无人 · 夹紧螺钉按 20/40/60 ft-lbs 逐级紧固

作业步骤	· 用千分表检查两个耦合标记点。 · 超差时拆 plug screw 并松开夹紧块螺钉。 · 旋转耦合头使差值 < 0.0008 in。 · 按 20、40、60 ft-lbs逐级紧固并复测。 · 复装 plug screw、动力刀具和前盖。
验证方式	动力刀具完全啮合，差值 < 0.0008 in。
验收标准	动力刀具完全啮合，差值 < 0.0008 in。
SOP编号	SOP-TC820LTYsi-315
程序名称	动力刀具驱动对中
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	TC820LTYsi 动力刀具驱动对中：按手册页码执行并以验收条件确认结果。
关键词	· BMT45 · 0.0008 in · 60 ft-lbs
检索文本	TC820LTYsi; sop; sop; live tooling; 动力刀具驱动对中; TC820LTYsi 动力刀具驱动对中：按手册页码执行并以验收条件确认结果。；用千分表检查两个耦合标记点。；超差时拆 plug screw 并松开夹紧块螺钉。；旋转耦合头使差值 < 0.0008 in。；按 20、40、60 ft-lbs逐级紧固并复测。；复装 plug screw、动力刀具和前盖。；BMT45; 0.0008 in; 60 ft-lbs; If the driven tools are not properly engaging with the driven tooling drive check turret to live tooling drive alignment: • Remove all driven tools and the front cover of 5.2.2.5a BMT45 Live Tooling Gear Head the turret. Alignment Markers • Take the machine out of E-Stop and with an 5.2.2.5b BMT45 Plug Screw indicator check the alignment of the 2 marking points on the live tooling gear head coupling 5.2.2.5c BMT45 Gear Head Clamping Screw along the X-Axis, shown in figure 5.2.2.5a. • The level of the 2 marking points along the X- Axis direction should be within 0.0008in or 0.2mm. If the level is greater than 0.0008” or 0.2mm along the X-Axis direction the gear head coupling will need to be adjusted. • If the level is out of spec remove the plug screw on the side of the turret housing, shown in figure 5.2.2.5b. • Loosen the screw of the shaft clamping block from the hole opened by removing the plug screw from the side of the turret housing, figure 5.2.2.5c. The Gear coupling head should now be able to rotate. o NOTE: This screw should be torqued to 60 ft-lbs. When tightening ensure it is tightened to 60 ft-lbs. • Rotate the gear head coupling and sweep the indicator checking the 2 markings on the gear head coupling until they are level across the X- Axis within 0.0008” or 0.2mm. • Fastening the clamping block screw to 20 ft-lbs then 40lbs and finally 60 ft-lbs checking the marking level remains less than 0.0008” or 0.2mm after each tighten. • When the clamping block screw is tightened to 60 ft-lbs and the 2 markings level remains less than 0.0008” or 0.2mm reinstall the plug screw on the side of the turret housing. • Install a driven tool and verify the drive is fully engaged. • Reinstall the front cover of the tool disc and tighten the M6 bolts to 14 ft-lbs.

原文摘录

If the driven tools are not properly engaging with the driven tooling drive check turret to live tooling drive alignment: • Remove all driven tools and the front cover of 5.2.2.5a BMT45 Live Tooling Gear Head the turret. Alignment Markers • Take the machine out of E-Stop and with an 5.2.2.5b BMT45 Plug Screw indicator check the alignment of the 2 marking points on the live tooling gear head coupling 5.2.2.5c BMT45 Gear Head Clamping Screw along the X-Axis, shown in figure 5.2.2.5a. • The level of the 2 marking points along the X- Axis direction should be within 0.0008in or 0.2mm. If the level is greater than 0.0008” or 0.2mm along the X-Axis direction the gear head coupling will need to be adjusted. • If the level is out of spec remove the plug screw on the side of the turret housing, shown in figure 5.2.2.5b. • Loosen the screw of the shaft clamping block from the hole opened by removing the plug screw from the side of the turret housing, figure 5.2.2.5c. The Gear coupling head should now be able to rotate. o NOTE: This screw should be torqued to 60 ft-lbs. When tightening ensure it is tightened to 60 ft-lbs. • Rotate the gear head coupling and sweep the indicator checking the 2 markings on the gear head coupling until they are level across the X- Axis within 0.0008” or 0.2mm. • Fastening the clamping block screw to 20 ft-lbs then 40lbs and finally 60 ft-lbs checking the marking level remains less than 0.0008” or 0.2mm after each tighten. • When the clamping block screw is tightened to 60 ft-lbs and the 2 markings level remains less than 0.0008” or 0.2mm reinstall the plug screw on the side of the turret housing. • Install a driven tool and verify the drive is fully engaged. • Reinstall the front cover of the tool disc and tighten the M6 bolts to 14 ft-lbs.

#19 lty-sop-probe-table			SOP 作业程序
SOP编号	SOP-TC820LTYsi-316	程序名称	探头校准前的刀具表设置
部件	automatic tool setter probe	章节	TC820LTYsi Model SOP
起始页	122	安全等级	high
知识类型	sop	适用机型	TC820LTYsi

摘要 | TC820LTYsi 探头校准前的刀具表设置：按手册页码执行并以验收条件确认结果。

A • 标识与来源

记录ID	lty-sop-probe-table
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	TC820LTYsi
型号范围	TC820LTYsi
型号专属	是
章节编号	LTY-SOP
章节	TC820LTYsi Model SOP
起始页	122

结束页	123
C • 分类与治理	
记录类别	sop
知识类型	sop
部件	automatic tool setter probe
安全等级	high
可入库	是
需人工复核	是

D • 业务明细

前置条件	· 准备 1/3/5 号校准刀具 · 相邻刀具无干涉
安全警告	· 清除工作偏置后建立机器零点 · 确认刀具不会撞击探头/卡盘
作业步骤	· 工位 1 装 OD 粗车刀，3 装 ID 粗车刀，5 装切断刀。 · 在 Tool List 新建并填写刀具数据。 · 从 T,S,M 清除工作偏置，选择 G500。
验证方式	三把刀具数据完整且 G500 生效。
验收标准	三把刀具数据完整且 G500 生效。
SOP编号	SOP-TC820LTYsi-316
程序名称	探头校准前的刀具表设置
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	TC820LTYsi 探头校准前的刀具表设置：按手册页码执行并以验收条件确认结果。
关键词	· Tool List · G500 · 工位1/3/5

检索文本

TC820LTYsi; sop; sop; automatic tool setter probe; 探头校准前的刀具表设置; TC820LTYsi 探头校准前的刀具表设置: 按手册页码执行并以验收条件确认结果。; 工位 1 装 OD 粗车刀, 3 装 ID 粗车刀, 5 装切断刀。; 在 Tool List 新建并填写刀具数据。; 从 T,S,M 清除工作偏置, 选择 G500。; Tool List; G500; 工位1/3/5; 8.0 Tool Setter Probe 8.1 About Probe 8.1.1 Manual Tool Setting Arm and Probe A manual tool setting arm and probe is fitted to the TC820si machine as standard. The tool setting arm is manually deployed by the user and can be used in the jog environment to set tool data. The position of the tool setting arm and probe are stored in the machine and used to set the position of the tools cutting edge with reference to the work piece and machine coordinate system. If the tool setting arm or probe is ever crashed, it will require recalibration. Although your probe will be calibrated at the factory before shipping it is suggested that you recalibrate the probe before using it to set any tools. Figure 7.1.1 Manual Tool Setting Arm and Probe 8.1.2 Automatic Tool Setting Arm and Probe An automatic tool setting arm and probe is fitted to the TC820LTYsi machine as standard. The tool setting arm is automatically deployed by the button on the MCP or with M-Codes. The automatic probe can be used in the jog environment to set tool data or in auto to check for tools throughout a part program or production run. The position of the tool setting arm and probe are stored in the machine and used to set the position of the tools cutting edge with reference to the work piece and machine coordinate system. If the tool setting arm or probe is ever crashed, it will require recalibration. Although your probe will be calibrated at the factory before shipping it is suggested that you recalibrate the probe before using it to set any tools. Figure 7.1.2 Automatic Tool Setting Arm and Probe 8.2 Tool Probe Calibration It is recommended you calibrate your tool setting arm and probe as soon as you receive your TCsi machine from the factory. The probe will also need to be recalibrated if a collision occurs. 8.2.1 Setting Up the Tool Table In order to calibrate all 4 sides of the tool setting probe 3 different tools will need to be used. To calibrate the tool setting probe, set up: • An OD roughing tool in position #1 on the turret. • An ID roughing tool in position #3 on the turret. • A cutoff tool in position #5 on the turret Populate your tool table with the information for the 3 tools installed on the turret following the instructions below, see figure 7.2a for a sample of the completed Tool list:

原文摘录

8.0 Tool Setter Probe 8.1 About Probe 8.1.1 Manual Tool Setting Arm and Probe A manual tool setting arm and probe is fitted to the TC820si machine as standard. The tool setting arm is manually deployed by the user and can be used in the jog environment to set tool data. The position of the tool setting arm and probe are stored in the machine and used to set the position of the tools cutting edge with reference to the work piece and machine coordinate system. If the tool setting arm or probe is ever crashed, it will require recalibration. Although your probe will be calibrated at the factory before shipping it is suggested that you recalibrate the probe before using it to set any tools. Figure 7.1.1 Manual Tool Setting Arm and Probe 8.1.2 Automatic Tool Setting Arm and Probe An automatic tool setting arm and probe is fitted to the TC820LTYsi machine as standard. The tool setting arm is automatically deployed by the button on the MCP or with M-Codes. The automatic probe can be used in the jog environment to set tool data or in auto to check for tools throughout a part program or production run. The position of the tool setting arm and probe are stored in the machine and used to set the position of the tools cutting edge with reference to the work piece and machine coordinate system. If the tool setting arm or probe is ever crashed, it will require recalibration. Although your probe will be calibrated at the factory before shipping it is suggested that you recalibrate the probe before using it to set any tools. Figure 7.1.2 Automatic Tool Setting Arm and Probe 8.2 Tool Probe Calibration It is recommended you calibrate your tool setting arm and probe as soon as you receive your TCsi machine from the factory. The probe will also need to be recalibrated if a collision occurs. 8.2.1 Setting Up the Tool Table In order to calibrate all 4 sides of the tool setting probe 3 different tools will need to be used. To calibrate the tool setting probe, set up: • An OD roughing tool in position #1 on the turret. • An ID roughing tool in position #3 on the turret. • A cutoff tool in position #5 on the turret Populate your tool table with the information for the 3 tools installed on the turret following the instructions below, see figure 7.2a for a sample of the completed Tool list:

#20 lty-sop-probe-calibration			SOP 作业程序
SOP编号	SOP-TC820LTYsi-317	程序名称	自动刀具设定臂和探头校准
部件	automatic tool setter probe	章节	TC820LTYsi Model SOP
起始页	125	安全等级	critical
知识类型	sop	适用机型	TC820LTYsi

摘要 | TC820LTYsi 自动刀具设定臂和探头校准：按手册页码执行并以验收条件确认结果。

A • 标识与来源

记录ID	lty-sop-probe-calibration
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	TC820LTYsi
型号范围	TC820LTYsi
型号专属	是
章节编号	LTY-SOP
章节	TC820LTYsi Model SOP
起始页	125
结束页	126

C • 分类与治理

记录类别	sop
知识类型	sop
部件	automatic tool setter probe
安全等级	critical
可入库	是
需人工复核	是

D • 业务明细

前置条件	· 刀具表和 G500 已准备 · 门互锁正常
安全警告	· 自动探头只能门关闭时部署 · 起始距离约 0.5 in 且无干涉

作业步骤	· 选择 Meas. Tool → Calibrate。 · 用 MCP 部署自动刀具设定臂。 · 将适用刀具移至所选探头边缘约 0.5 in。 · Cycle Start 触碰并更新数据。 · 对四个探头面重复。
验证方式	四面数据更新且无碰撞/报警。
验收标准	四面数据更新且无碰撞/报警。
SOP编号	SOP-TC820LTYsi-317
程序名称	自动刀具设定臂和探头校准
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	TC820LTYsi 自动刀具设定臂和探头校准：按手册页码执行并以验收条件确认结果。
关键词	· 四面校准 · 0.5 in · 探头
检索文本	TC820LTYsi; sop; sop; automatic tool setter probe; 自动刀具设定臂和探头校准; TC820LTYsi 自动刀具设定臂和探头校准：按手册页码执行并以验收条件确认结果。；选择 Meas. Tool → Calibrate。；用 MCP 部署自动刀具设定臂。；将适用刀具移至所选探头边缘约 0.5 in。；Cycle Start 触碰并更新数据。；对四个探头面重复。；四面校准；0.5 in；探头；8.2.3 Calibrate the Tool Setting Arm and Probe To calibrate the tool setting arm and probe press [M] from the left VSK. Make sure any adjacent tools will not hit the probe or chuck during the calibration process. Press “Meas. Tool” from the HSK and “Calibrate” from the right VSK. Press one of the buttons on the right VSK for the direction to be calibrated shown in figure 7.2.3a. Figure 7.2.3a Probe Calibration Screen • On the TC820si for the manual tool setter there is a release knob inside of the machine envelope just under the chuck. Pull the knob as shown in figure 7.2.3b. Swing the probe arm into position as shown in figure 7.2.3c. There is a light on the probe to show it is active, this only works when the door is closed. Figure 7.2.3b Release Knob Figure 7.2.3c Probe in Position
原文摘录	8.2.3 Calibrate the Tool Setting Arm and Probe To calibrate the tool setting arm and probe press [M] from the left VSK. Make sure any adjacent tools will not hit the probe or chuck during the calibration process. Press “Meas. Tool” from the HSK and “Calibrate” from the right VSK. Press one of the buttons on the right VSK for the direction to be calibrated shown in figure 7.2.3a. Figure 7.2.3a Probe Calibration Screen • On the TC820si for the manual tool setter there is a release knob inside of the machine envelope just under the chuck. Pull the knob as shown in figure 7.2.3b. Swing the probe arm into position as shown in figure 7.2.3c. There is a light on the probe to show it is active, this only works when the door is closed. Figure 7.2.3b Release Knob Figure 7.2.3c Probe in Position

#21 lty-sop-probe-jog			SOP 作业程序
SOP编号	SOP-TC820LTYsi-318	程序名称	JOG 模式探头测量刀具
部件	automatic tool setter probe	章节	TC820LTYsi Model SOP
起始页	126	安全等级	high
知识类型	sop	适用机型	TC820LTYsi

摘要 | TC820LTYsi JOG 模式探头测量刀具：按手册页码执行并以验收条件确认结果。

A • 标识与来源

记录ID	lty-sop-probe-jog
文档ID	trak-tcsi-service-34242-v022725

文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	TC820LTYsi
型号范围	TC820LTYsi
型号专属	是
章节编号	LTY-SOP
章节	TC820LTYsi Model SOP
起始页	126
结束页	127

C • 分类与治理

记录类别	sop
知识类型	sop
部件	automatic tool setter probe
安全等级	high
可入库	是
需人工复核	是

D • 业务明细

前置条件	· 探头已校准 · 相邻刀具无干涉
安全警告	· MPG 低速靠近并确认路径 · 异常立即停止
作业步骤	· 选择 Meas. Tool → Automatic。 · 选择方向并查看提示。 · 用 MPG 移到探头前约 1/2 in。 · Cycle Start 完成接触并重复另一轴。 · 确认 Tool data 更新。
验证方式	目标刀具 X/Z 数据更新。
验收标准	目标刀具 X/Z 数据更新。
SOP编号	SOP-TC820LTYsi-318
程序名称	JOG 模式探头测量刀具
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	TC820LTYsi JOG 模式探头测量刀具：按手册页码执行并以验收条件确认结果。
------	--

关键词	· JOG · MPG · Tool data
检索文本	TC820LTYsi; sop; sop; automatic tool setter probe; JOG 模式探头测量刀具; TC820LTYsi JOG 模式探头测量刀具: 按手册页码执行并以验收条件确认结果。; 选择 Meas. Tool → Automatic。; 选择方向并查看提示。; 用 MPG 移到探头前约 1/2 in。; Cycle Start 完成接触并重复另一轴。; 确认 Tool data 更新。; JOG; MPG; Tool data; • On the TC820LTYsi for the automatic tool setter the tool setting arm is automatically deployed by pressing the button on the MCP, figure 7.2.3d, or through the use of M-Codes. The arm will swing into position as shown in figure 7.2.3e. The probe can only be deployed when the door is closed. Figure 7.2.3d Probe Deploy Button Figure 7.2.3e Probe in Position All 4 sides of the tool setter will need to be calibrated, move the tool to within 0.5" of the edge you selected on the calibration screen on the machine control, figure 7.2.3a, and press cycle start. The tool should move and touch the probe updating the probe data. Repeat this process for all 4 sides of the probe using the appropriate tool that can reach and touch off the desired side. 8.3 Measuring Tools Using the Tool Setting Arm and Probe in Jog Press [M] from the left VSK. Make sure any adjacent tools will not hit the probe or chuck during the measurement cycle. Press "Meas. Tool" from the HSK and "Automatic" from the right VSK. Press one of the buttons on the right VSK for the direction to be measured. The screen will display which of the four sides of the probe need to be touched by the tool as shown in figure 7.3a. Use MPG to move the active tool about ½" away from the correct probe. Press cycle start. Figure 7.3a First Position
原文摘录	• On the TC820LTYsi for the automatic tool setter the tool setting arm is automatically deployed by pressing the button on the MCP, figure 7.2.3d, or through the use of M-Codes. The arm will swing into position as shown in figure 7.2.3e. The probe can only be deployed when the door is closed. Figure 7.2.3d Probe Deploy Button Figure 7.2.3e Probe in Position All 4 sides of the tool setter will need to be calibrated, move the tool to within 0.5" of the edge you selected on the calibration screen on the machine control, figure 7.2.3a, and press cycle start. The tool should move and touch the probe updating the probe data. Repeat this process for all 4 sides of the probe using the appropriate tool that can reach and touch off the desired side. 8.3 Measuring Tools Using the Tool Setting Arm and Probe in Jog Press [M] from the left VSK. Make sure any adjacent tools will not hit the probe or chuck during the measurement cycle. Press "Meas. Tool" from the HSK and "Automatic" from the right VSK. Press one of the buttons on the right VSK for the direction to be measured. The screen will display which of the four sides of the probe need to be touched by the tool as shown in figure 7.3a. Use MPG to move the active tool about ½" away from the correct probe. Press cycle start. Figure 7.3a First Position

#22 lty-sop-probe-auto			SOP 作业程序
SOP编号	SOP-TC820LTYsi-319	程序名称	AUTO 模式探头测量刀具
部件	automatic tool setter probe	章节	TC820LTYsi Model SOP
起始页	127	安全等级	high
知识类型	sop	适用机型	TC820LTYsi

摘要 | TC820LTYsi AUTO 模式探头测量刀具: 按手册页码执行并以验收条件确认结果。

A • 标识与来源

记录ID	lty-sop-probe-auto
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.

文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	TC820LTYsi
型号范围	TC820LTYsi
型号专属	是
章节编号	LTY-SOP
章节	TC820LTYsi Model SOP
起始页	127
结束页	127

C • 分类与治理

记录类别	sop
知识类型	sop
部件	automatic tool setter probe
安全等级	high
可入库	是
需人工复核	是

D • 业务明细

前置条件	· 探头已校准 · 零件程序已验证
安全警告	· 嵌入生产程序前验证所有干涉 · 报警时停止程序检查反馈
作业步骤	· 在 Auto 调用 Siemens 测量循环。 · 核对刀塔布置和相邻工具间隙。 · 单件低风险试运行。 · 确认结果写入正确刀具补偿。
验证方式	Auto 测量循环无碰撞和报警。
验收标准	Auto 测量循环无碰撞和报警。
SOP编号	SOP-TC820LTYsi-319
程序名称	AUTO 模式探头测量刀具
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	TC820LTYsi AUTO 模式探头测量刀具：按手册页码执行并以验收条件确认结果。
关键词	· AUTO · 测量循环 · 刀具补偿

检索文本

TC820LTYsi; sop; sop; automatic tool setter probe; AUTO 模式探头测量刀具; TC820LTYsi AUTO 模式探头测量刀具: 按手册页码执行并以验收条件确认结果。; 在 Auto 调用 Siemens 测量循环。; 核对刀塔布置和相邻工具间隙。; 单件低风险试运行。; 确认结果写入正确刀具补偿。; AUTO; 测量循环; 刀具补偿; Repeat this process for the other axis displayed on the right VSK as shown in figure 7.3b. Figure 7.3b Second Position This process will fill in the dimensions in the “Tool data” chart in the Work Area. Repeat this same process using whichever tool needs to be measured. 8.4 Measuring Tools Using the Tool Setting Arm and Probe in Auto The TC820LTY automatic tool setting arm and probe with the Siemens measuring cycle software can be used in the Auto environment and embedded in a part program and production run. Before the probe measuring cycle can be embedded in a program the tool setting arm and probe needs to be calibrated. Pay attention to the tooling set up on the turret so any tool being measured will not have other tools interfere and hit the tool setting arm or the chuck. 8.5 Tool Setting Probe Troubleshooting The probe is designed to turn on when the tool setting arm is deployed. When the probe signal is not working for the TC820si manual tool setting arm and probe: • Check the door is closed and the interlock is engaged. The door open message on the control should not be showing. • Check the connecting cable in the X222 area of the TC820si electrical cabinet and make sure the red wire is connected properly to X222-216-1 and the white wire is connected properly to X132.13- 8. • There are limit switches that show when the probe is deployed and parked. Refer to the TRAK I/O Diagnostic screen and make sure all the inputs and outputs for the probe and tool setting arm limit switches are working as expected. When the probe signal is not working for the TC820LTYsi automatic tool setting arm and probe: • Check the door is closed and the interlock is engaged. The door open message on the control should not be showing.

原文摘录

Repeat this process for the other axis displayed on the right VSK as shown in figure 7.3b. Figure 7.3b Second Position This process will fill in the dimensions in the “Tool data” chart in the Work Area. Repeat this same process using whichever tool needs to be measured. 8.4 Measuring Tools Using the Tool Setting Arm and Probe in Auto The TC820LTY automatic tool setting arm and probe with the Siemens measuring cycle software can be used in the Auto environment and embedded in a part program and production run. Before the probe measuring cycle can be embedded in a program the tool setting arm and probe needs to be calibrated. Pay attention to the tooling set up on the turret so any tool being measured will not have other tools interfere and hit the tool setting arm or the chuck. 8.5 Tool Setting Probe Troubleshooting The probe is designed to turn on when the tool setting arm is deployed. When the probe signal is not working for the TC820si manual tool setting arm and probe: • Check the door is closed and the interlock is engaged. The door open message on the control should not be showing. • Check the connecting cable in the X222 area of the TC820si electrical cabinet and make sure the red wire is connected properly to X222-216-1 and the white wire is connected properly to X132.13- 8. • There are limit switches that show when the probe is deployed and parked. Refer to the TRAK I/O Diagnostic screen and make sure all the inputs and outputs for the probe and tool setting arm limit switches are working as expected. When the probe signal is not working for the TC820LTYsi automatic tool setting arm and probe: • Check the door is closed and the interlock is engaged. The door open message on the control should not be showing.

#23 lty-sop-probe-fault		SOP 作业程序	
SOP编号	SOP-TC820LTYsi-320	程序名称	自动探头信号故障检查
部件	automatic tool setter probe	章节	TC820LTYsi Model SOP
起始页	127	安全等级	high
知识类型	sop	适用机型	TC820LTYsi

摘要 | TC820LTYsi 自动探头信号故障检查: 按手册页码执行并以验收条件确认结果。

A • 标识与来源

记录ID	lty-sop-probe-fault
文档ID	trak-tcsi-service-34242-v022725

文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	TC820LTYsi
型号范围	TC820LTYsi
型号专属	是
章节编号	LTY-SOP
章节	TC820LTYsi Model SOP
起始页	127
结束页	128

C • 分类与治理

记录类别	sop
知识类型	sop
部件	automatic tool setter probe
安全等级	high
可入库	是
需人工复核	是

D • 业务明细

前置条件	探头信号异常且机床停止相关动作
安全警告	· 不得带电插拔电柜连接器 · 门互锁异常不得旁路运行
作业步骤	· 确认门关闭且互锁有效。 · 检查 TSI 3 到 X111、X222、TB4 的电缆。 · 在 TRAK I/O Diagnostic 观察限位输入输出。 · 参考 Renishaw 手册并保存报警/I/O证据。
验证方式	部署、停车和触发反馈与实际状态一致。
验收标准	部署、停车和触发反馈与实际状态一致。
SOP编号	SOP-TC820LTYsi-320
程序名称	自动探头信号故障检查
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	TC820LTYsi 自动探头信号故障检查：按手册页码执行并以验收条件确认结果。
------	--

关键词	· TSI 3 · X111 · X222 · TB4
检索文本	TC820LTYsi; sop; sop; automatic tool setter probe; 自动探头信号故障检查; TC820LTYsi 自动探头信号故障检查: 按手册页码执行并以验收条件确认结果。; 确认门关闭且互锁有效。; 检查 TSI 3 到 X111、X222、TB4 的电缆。; 在 TRAK I/O Diagnostic 观察限位输入输出。; 参考 Renishaw 手册并保存报警/I/O证据。; TSI 3; X111; X222; TB4; Repeat this process for the other axis displayed on the right VSK as shown in figure 7.3b. Figure 7.3b Second Position This process will fill in the dimensions in the “Tool data” chart in the Work Area. Repeat this same process using whichever tool needs to be measured. 8.4 Measuring Tools Using the Tool Setting Arm and Probe in Auto The TC820LTY automatic tool setting arm and probe with the Siemens measuring cycle software can be used in the Auto environment and embedded in a part program and production run. Before the probe measuring cycle can be embedded in a program the tool setting arm and probe needs to be calibrated. Pay attention to the tooling set up on the turret so any tool being measured will not have other tools interfere and hit the tool setting arm or the chuck. 8.5 Tool Setting Probe Troubleshooting The probe is designed to turn on when the tool setting arm is deployed. When the probe signal is not working for the TC820si manual tool setting arm and probe: • Check the door is closed and the interlock is engaged. The door open message on the control should not be showing. • Check the connecting cable in the X222 area of the TC820si electrical cabinet and make sure the red wire is connected properly to X222-216-1 and the white wire is connected properly to X132.13- 8. • There are limit switches that show when the probe is deployed and parked. Refer to the TRAK I/O Diagnostic screen and make sure all the inputs and outputs for the probe and tool setting arm limit switches are working as expected. When the probe signal is not working for the TC820LTYsi automatic tool setting arm and probe: • Check the door is closed and the interlock is engaged. The door open message on the control should not be showing.
原文摘录	Repeat this process for the other axis displayed on the right VSK as shown in figure 7.3b. Figure 7.3b Second Position This process will fill in the dimensions in the “Tool data” chart in the Work Area. Repeat this same process using whichever tool needs to be measured. 8.4 Measuring Tools Using the Tool Setting Arm and Probe in Auto The TC820LTY automatic tool setting arm and probe with the Siemens measuring cycle software can be used in the Auto environment and embedded in a part program and production run. Before the probe measuring cycle can be embedded in a program the tool setting arm and probe needs to be calibrated. Pay attention to the tooling set up on the turret so any tool being measured will not have other tools interfere and hit the tool setting arm or the chuck. 8.5 Tool Setting Probe Troubleshooting The probe is designed to turn on when the tool setting arm is deployed. When the probe signal is not working for the TC820si manual tool setting arm and probe: • Check the door is closed and the interlock is engaged. The door open message on the control should not be showing. • Check the connecting cable in the X222 area of the TC820si electrical cabinet and make sure the red wire is connected properly to X222-216-1 and the white wire is connected properly to X132.13- 8. • There are limit switches that show when the probe is deployed and parked. Refer to the TRAK I/O Diagnostic screen and make sure all the inputs and outputs for the probe and tool setting arm limit switches are working as expected. When the probe signal is not working for the TC820LTYsi automatic tool setting arm and probe: • Check the door is closed and the interlock is engaged. The door open message on the control should not be showing.

#24 lty-sop-cabinet-filter			SOP 作业程序
SOP编号	SOP-TC820LTYsi-321	程序名称	电柜热交换器过滤器清洁
部件	electrical cabinet cooling	章节	TC820LTYsi Model SOP
起始页	84	安全等级	high
知识类型	sop	适用机型	TC820LTYsi

摘要 | TC820LTYsi 电柜热交换器过滤器清洁: 按手册页码执行并以验收条件确认结果。

A • 标识与来源

记录ID	lty-sop-cabinet-filter
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	TC820LTYsi
型号范围	TC820LTYsi
型号专属	是
章节编号	LTY-SOP
章节	TC820LTYsi Model SOP
起始页	84
结束页	84

C • 分类与治理

记录类别	sop
知识类型	sop
部件	electrical cabinet cooling
安全等级	high
可入库	是
需人工复核	是

D • 业务明细

前置条件	· 机床停机并断电 · 过滤器可安全取出
安全警告	· 过滤器干燥后再装回 · 不得带电接触电柜内部
作业步骤	· 抬底部卡舌取出过滤器。 · 用空气吹净或水洗。 · 清洁风扇格栅散热片。 · 干燥后滑回槽内固定。
验证方式	过滤器和冷却通道无堵塞。
验收标准	过滤器和冷却通道无堵塞。
SOP编号	SOP-TC820LTYsi-321
程序名称	电柜热交换器过滤器清洁
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	TC820LTYsi 电柜热交换器过滤器清洁：按手册页码执行并以验收条件确认结果。
关键词	· 电柜 · 热交换器 · 过滤器
检索文本	TC820LTYsi; sop; sop; electrical cabinet cooling; 电柜热交换器过滤器清洁；TC820LTYsi 电柜热交换器过滤器清洁：按手册页码执行并以验收条件确认结果。；抬底部卡舌取出过滤器。；用空气吹净或水洗。；清洁风扇格栅散热片。；干燥后滑回槽内固定。；电柜；热交换器；过滤器；5.0 Maintenance and Replacement 5.1 Maintenance 5.1.1 How to Clean the Touchscreen & Machine Control Panel (MCP) The pendant is designed for low maintenance operation. Nevertheless, you must regularly clean the pendant. WARNING!!! 1. Turn power off when cleaning the pendant. 2. Do not clean the device using compressed air or steam jets because they can damage it. Clean only the front panel with a soft cloth moistened with water of mild cleaning agent. Spray the cleaning agent or water on to a cleaning cloth. Do not spray anything directly on the front panel. Never use caustic solvents or abrasive cleaners. Use dry cloth to clean the backside of the pendant. 5.1.2 How to Clean the Electrical Cabinet Heat Exchanger Air Filter 1. Lift tab at the bottom of the filter which lifts the filter out of the slot in front of the fan, see figure 4.1.2. 2. Clean filter either by blowing with air or rinsing with water. 3. Clean fins on the grate of the fan. 4. Dry filter if needed and replace into the slot and slide down to secure. Figure 4.1.2 Heat Exchanger Filter
原文摘录	5.0 Maintenance and Replacement 5.1 Maintenance 5.1.1 How to Clean the Touchscreen & Machine Control Panel (MCP) The pendant is designed for low maintenance operation. Nevertheless, you must regularly clean the pendant. WARNING!!! 1. Turn power off when cleaning the pendant. 2. Do not clean the device using compressed air or steam jets because they can damage it. Clean only the front panel with a soft cloth moistened with water of mild cleaning agent. Spray the cleaning agent or water on to a cleaning cloth. Do not spray anything directly on the front panel. Never use caustic solvents or abrasive cleaners. Use dry cloth to clean the backside of the pendant. 5.1.2 How to Clean the Electrical Cabinet Heat Exchanger Air Filter 1. Lift tab at the bottom of the filter which lifts the filter out of the slot in front of the fan, see figure 4.1.2. 2. Clean filter either by blowing with air or rinsing with water. 3. Clean fins on the grate of the fan. 4. Dry filter if needed and replace into the slot and slide down to secure. Figure 4.1.2 Heat Exchanger Filter

#25 lty-sop-hydraulic-filter			SOP 作业程序
SOP编号	SOP-TC820LTYsi-322	程序名称	液压泵过滤器更换
部件	hydraulic system	章节	TC820LTYsi Model SOP
起始页	85	安全等级	critical
知识类型	sop	适用机型	TC820LTYsi

摘要 | TC820LTYsi 液压泵过滤器更换：按手册页码执行并以验收条件确认结果。

A • 标识与来源

记录ID	lty-sop-hydraulic-filter
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	TC820LTYsi
型号范围	TC820LTYsi
型号专属	是
章节编号	LTY-SOP
章节	TC820LTYsi Model SOP
起始页	85
结束页	85

C • 分类与治理

记录类别	sop
知识类型	sop
部件	hydraulic system
安全等级	critical
可入库	是
需人工复核	是

D • 业务明细

前置条件	· 系统断电泄压 · 准备起吊设备和新过滤器
安全警告	· 油箱上盖是重物必须起吊 · 防止油液喷出和污染
作业步骤	· 拆六个螺钉和两个吊环螺栓。 · 用起吊设备分离并提升上盖。 · 更换过滤器。 · 复位上盖并固定油箱。
验证方式	过滤器已换、密封正常、无泄漏。
验收标准	过滤器已换、密封正常、无泄漏。
SOP编号	SOP-TC820LTYsi-322
程序名称	液压泵过滤器更换
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	TC820LTYsi 液压泵过滤器更换：按手册页码执行并以验收条件确认结果。
关键词	· 液压泵 · 过滤器 · 起吊设备

检索文本

TC820LTYsi; sop; sop; hydraulic system; 液压泵过滤器更换; TC820LTYsi 液压泵过滤器更换: 按手册页码执行并以验收条件确认结果。; 拆六个螺钉和两个吊环螺栓。; 用起吊设备分离并提升上盖。; 更换过滤器。; 复位上盖并固定油箱。; 液压泵; 过滤器; 起吊设备; 5.1.3 How to Replace Hydraulic Pump Filter WARNING!!! Heavy Object, us proper lifting equipment. 1. Remove the six screws and two eye bolts between the top cover and the tank, see figure 4.1.3. 2. Use proper lifting equipment to separate the top cover from the tank. 3. Lift the top cover high enough to replace the filter. 4. Replace top cover and secure to tank. Figure 4.1.3 Hydraulic Pump 5.2 Maintenance Intervals 5.2.1 Daily Maintenance • Remove majority of chips and debris from around the ways, way covers, chuck, turret, tool holders and other moving parts. • Empty chips from screen sitting above the coolant tank. • Check hydraulic oil levels. If the oil level is low check for leaks in the hydraulic system. If needed fill with ISO VG32/SAE 10W. • Check the main hydraulic pressure is set to 700psi. • Check the lubrication pump oil level and make sure it is always above the minimum line. If needed fill with ISO VG68/SAE 20W. • Check to make sure there are no obvious oil leaks to the lubrication system, and it is getting to all the slideways and moving parts. • Check lubrication pressure is set to 70psi when the pump is running. • Check to make sure the clamping mechanism is clean and in proper working order. Grease the chuck twice daily. • Check coolant level and add if low.

原文摘录

5.1.3 How to Replace Hydraulic Pump Filter WARNING!!! Heavy Object, us proper lifting equipment. 1. Remove the six screws and two eye bolts between the top cover and the tank, see figure 4.1.3. 2. Use proper lifting equipment to separate the top cover from the tank. 3. Lift the top cover high enough to replace the filter. 4. Replace top cover and secure to tank. Figure 4.1.3 Hydraulic Pump 5.2 Maintenance Intervals 5.2.1 Daily Maintenance • Remove majority of chips and debris from around the ways, way covers, chuck, turret, tool holders and other moving parts. • Empty chips from screen sitting above the coolant tank. • Check hydraulic oil levels. If the oil level is low check for leaks in the hydraulic system. If needed fill with ISO VG32/SAE 10W. • Check the main hydraulic pressure is set to 700psi. • Check the lubrication pump oil level and make sure it is always above the minimum line. If needed fill with ISO VG68/SAE 20W. • Check to make sure there are no obvious oil leaks to the lubrication system, and it is getting to all the slideways and moving parts. • Check lubrication pressure is set to 70psi when the pump is running. • Check to make sure the clamping mechanism is clean and in proper working order. Grease the chuck twice daily. • Check coolant level and add if low.

#26 lty-sop-gibs		SOP 作业程序	
SOP编号	SOP-TC820LTYsi-323	程序名称	各轴和尾座镶条调整
部件	gibs	章节	TC820LTYsi Model SOP
起始页	86	安全等级	high
知识类型	sop	适用机型	TC820LTYsi

摘要 | TC820LTYsi 各轴和尾座镶条调整: 按手册页码执行并以验收条件确认结果。

A • 标识与来源

记录ID	lty-sop-gibs
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	TC820LTYsi
型号范围	TC820LTYsi
型号专属	是
章节编号	LTY-SOP
章节	TC820LTYsi Model SOP
起始页	86
结束页	92

C • 分类与治理

记录类别	sop
知识类型	sop
部件	gibs
安全等级	high
可入库	是
需人工复核	是

D • 业务明细

前置条件	· 轴停止且区域清洁 · 确认对应轴和镶条类型
安全警告	· 错误调整会损失精度 · 禁止用垫片替代，避免气流吹屑入滑道
作业步骤	· 检查镶条平直、无毛刺且油道畅通。 · TC820LTYsi X/Y/Z 从后盖、B 轴从尾座防护板进入。 · 用扭矩螺丝刀设为 1.2 Nm/10.62 lbf.in。 · 按单面/双面镶条步骤往复、复紧和锁定。 · 复装后检查运动、润滑和精度。
验证方式	运动平稳，调整扭矩和精度符合要求。
验收标准	运动平稳，调整扭矩和精度符合要求。
SOP编号	SOP-TC820LTYsi-323
程序名称	各轴和尾座镶条调整
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	TC820LTYsi 各轴和尾座镶条调整：按手册页码执行并以验收条件确认结果。
关键词	· X轴 · Y轴 · Z轴 · B轴 · 1.2 Nm

检索文本

TC820LTYsi; sop; sop; gibs; 各轴和尾座镶条调整; TC820LTYsi 各轴和尾座镶条调整: 按手册页码执行并以验收条件确认结果。; 检查镶条平直、无毛刺且油道畅通。; TC820LTYsi X/Y/Z 从后盖、B 轴从尾座防护板进入。; 用扭矩螺丝刀设为 1.2 Nm/10.62 lbf.in。; 按单面/双面镶条步骤往复、复紧和锁定。; 复装后检查运动、润滑和精度。; X轴; Y轴; Z轴; B轴; 1.2 Nm; 5.2.2 Monthly Maintenance • Clean air filters on the electrical cabinet heat exchanger and replace if needed. • Remove covers and inspect ways, clean areas of chips and debris. • Every 2 months remove and drain the coolant tank, clean inside and fill with new clean coolant. 5.2.3 Half Yearly Maintenance • Check all switches and interlocks are working correctly. • Check the level of the machine and re-level as necessary. • Check the spindle motor belt tension and adjust as necessary. • Check the backlash on each axis and adjust as necessary. • Check gibs on each axis and tailstock and adjust as necessary. 5.2.4 Yearly Maintenance • Remove the hydraulic tank. Drain oil and clean inside. Clean filter and fill with new clean hydraulic oil (ISO VG32/SAE 10W). • Remove all covers, vacuum chips and debris that may have built up, and wipe down machine from top to bottom. • Inspect machine for any unusual wear and play. • Check cables, coolant hoses and hydraulic lines for any abrasion or cuts. 5.3 Mechanical Adjustments 5.3.1 Gib Adjustments WARNING!!! Ensure to adjust gibs frequently and properly. Otherwise, this might result in loss of machine accuracy. The gibs are vital to the performance of your TCsi Machine. Gibs should be: • flat • free of twist • free of burrs • free of blockages in the oil passages and channels Defective or scarred gibs must be replaced. Shimming of gibs will not yield acceptable results. It is good machining practice to avoid the use of shop air to clean the chips off a machine. This risks blowing chips into the sliding way surfaces and compromising the performance of the machine. Because of long-term friction between the bed and carriage, wear may occur. Proper gib adjustment is necessary after the machine has been operated for a long time. Adjust the gib by turning the gib adjustment screw (2) using torque screwdriver see figure 4.3.1. Set torque to 1.2Nm (0.9 ft lbs).

原文摘录

5.2.2 Monthly Maintenance • Clean air filters on the electrical cabinet heat exchanger and replace if needed. • Remove covers and inspect ways, clean areas of chips and debris. • Every 2 months remove and drain the coolant tank, clean inside and fill with new clean coolant. 5.2.3 Half Yearly Maintenance • Check all switches and interlocks are working correctly. • Check the level of the machine and re-level as necessary. • Check the spindle motor belt tension and adjust as necessary. • Check the backlash on each axis and adjust as necessary. • Check gibs on each axis and tailstock and adjust as necessary. 5.2.4 Yearly Maintenance • Remove the hydraulic tank. Drain oil and clean inside. Clean filter and fill with new clean hydraulic oil (ISO VG32/SAE 10W). • Remove all covers, vacuum chips and debris that may have built up, and wipe down machine from top to bottom. • Inspect machine for any unusual wear and play. • Check cables, coolant hoses and hydraulic lines for any abrasion or cuts. 5.3 Mechanical Adjustments 5.3.1 Gib Adjustments WARNING!!! Ensure to adjust gibs frequently and properly. Otherwise, this might result in loss of machine accuracy. The gibs are vital to the performance of your TCsi Machine. Gibs should be: • flat • free of twist • free of burrs • free of blockages in the oil passages and channels Defective or scarred gibs must be replaced. Shimming of gibs will not yield acceptable results. It is good machining practice to avoid the use of shop air to clean the chips off a machine. This risks blowing chips into the sliding way surfaces and compromising the performance of the machine. Because of long-term friction between the bed and carriage, wear may occur. Proper gib adjustment is necessary after the machine has been operated for a long time. Adjust the gib by turning the gib adjustment screw (2) using torque screwdriver see figure 4.3.1. Set torque to 1.2Nm (0.9 ft lbs).

#27 lty-sop-spindle-belt		SOP 作业程序	
SOP编号	SOP-TC820LTYsi-324	程序名称	主轴皮带检查和重新张紧
部件	spindle belt	章节	TC820LTYsi Model SOP
起始页	94	安全等级	critical
知识类型	sop	适用机型	TC820LTYsi

摘要 | TC820LTYsi 主轴皮带检查和重新张紧: 按手册页码执行并以验收条件确认结果。

A • 标识与来源

记录IDlty-sop-spindle-belt

文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	TC820LTYsi
型号范围	TC820LTYsi
型号专属	是
章节编号	LTY-SOP
章节	TC820LTYsi Model SOP
起始页	94
结束页	96

C • 分类与治理

记录类别	sop
知识类型	sop
部件	spindle belt
安全等级	critical
可入库	是
需人工复核	是

D • 业务明细

前置条件	· 确认需检查/调整皮带 · 电源已断开并防止启动
安全警告	· 无法确保张力时不要调整 · 新皮带 48 小时后复查
作业步骤	· 松开电机板六个螺钉并调张紧器。 · TC820LTYsi 仪表输入 150 g/m、11 mm/#R、span 766 mm。 · 测量目标约 120 lbf，允许 105–145 lbf，新带约 145 lbf。 · 确认后六螺钉紧固 100 ft-lbs。
验证方式	张力、对中和扭矩符合要求，运行无异常噪声。
验收标准	张力、对中和扭矩符合要求，运行无异常噪声。
SOP编号	SOP-TC820LTYsi-324
程序名称	主轴皮带检查和重新张紧
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	TC820LTYsi 主轴皮带检查和重新张紧：按手册页码执行并以验收条件确认结果。
------	---

关键词	· 766 mm · 120 lbf · 105-145 lbf · 100 ft-lbs
检索文本	TC820LTYsi; sop; sop; spindle belt; 主轴皮带检查和重新张紧; TC820LTYsi 主轴皮带检查和重新张紧 : 按手册页码执行并以验收条件确认结果。; 松开电机板六个螺钉并调张紧器。; TC820LTYsi 仪表输入 150 g/m、11 mm/#R、span 766 mm。; 测量目标约 120 lbf, 允许 105–145 lbf, 新带约 145 lbf。; 确认后六螺钉紧固 100 ft-lbs。; 766 mm; 120 lbf; 105-145 lbf; 100 ft-lbs; 5.4.3 Check Spindle Belt WARNING!!! Ensure to have a proper tension value for the spindle transmission belt. If you cannot ensure the proper tension value, please do not adjust the belt tension. After the machine has been operated for a long period, the spindle drive-timing belt may gradually become loose. Check the main drive belt tension frequently. 5.4.4 Re-tension Spindle Belt Follow steps below to adjust the belt tension reference figure 4.4.2a. Two tool choices are explained to test the tension: 1. Make sure the power source has been disconnected before adjusting the timing belt tension. 2. Loosen the 6 fastening screws (1) on the spindle motor mounting plate. Figure 4.4.2a Spindle Belt Orientation 3. Set belt tension by adjusting the belt tension motor bracket reference figure 4.4.2b inset photo. Use the tensioner block to either increase tension on the belt or decrease tension. When the tension on the belt looks good, snug the 6 fastening screws.
原文摘录	5.4.3 Check Spindle Belt WARNING!!! Ensure to have a proper tension value for the spindle transmission belt. If you cannot ensure the proper tension value, please do not adjust the belt tension. After the machine has been operated for a long period, the spindle drive-timing belt may gradually become loose. Check the main drive belt tension frequently. 5.4.4 Re-tension Spindle Belt Follow steps below to adjust the belt tension reference figure 4.4.2a. Two tool choices are explained to test the tension: 1. Make sure the power source has been disconnected before adjusting the timing belt tension. 2. Loosen the 6 fastening screws (1) on the spindle motor mounting plate. Figure 4.4.2a Spindle Belt Orientation 3. Set belt tension by adjusting the belt tension motor bracket reference figure 4.4.2b inset photo. Use the tensioner block to either increase tension on the belt or decrease tension. When the tension on the belt looks good, snug the 6 fastening screws.

#28 lty-sop-backlash			SOP 作业程序
SOP编号	SOP-TC820LTYsi-325	程序名称	X/Z 轴反向间隙参数检查
部件	axis backlash	章节	TC820LTYsi Model SOP
起始页	97	安全等级	high
知识类型	sop	适用机型	TC820LTYsi

摘要 | TC820LTYsi X/Z 轴反向间隙参数检查：按手册页码执行并以验收条件确认结果。

A • 标识与来源

记录ID	lty-sop-backlash
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	TC820LTYsi
型号范围	TC820LTYsi
型号专属	是
章节编号	LTY-SOP
章节	TC820LTYsi Model SOP
起始页	97
结束页	97

C • 分类与治理

记录类别	sop
知识类型	sop
部件	axis backlash
安全等级	high
可入库	是
需人工复核	是

D • 业务明细

前置条件	· 确认重复性/定位精度需检查 · 具备制造商级参数权限
安全警告	· 32450[0] 是单机值，禁止套用其他机床 · Manufacturer level password 才可编辑
作业步骤	· 进入 Setup → Mach data → Axis MD。 · Search 参数 32450。 · Select axis 查看 AX1/AX2。 · 记录参数和测量结果，授权后才可调整。
验证方式	参数与单机实测和维修记录一致。
验收标准	参数与单机实测和维修记录一致。
SOP编号	SOP-TC820LTYsi-325
程序名称	X/Z 轴反向间隙参数检查
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	TC820LTYsi X/Z 轴反向间隙参数检查：按手册页码执行并以验收条件确认结果。
关键词	· 32450 · AX1 · AX2

检索文本	TC820LTYsi; sop; sop; axis backlash; X/Z 轴反向间隙参数检查; TC820LTYsi X/Z 轴反向间隙参数检查: 按手册页码执行并以验收条件确认结果。; 进入 Setup → Mach data → Axis MD。; Search 参数 32450。; Select axis 查看 AX1/AX2。; 记录参数和测量结果, 授权后才可调整。; 32450; AX1; AX2; 5.4.5 Backlash Constants Backlash settings for the TCsi machines are set by the factory during the build for the X & Z-Axis. These values are unique for each machine. The backlash values for the two axes, AX1 (X-Axis) and AX2 (Z-Axis) are stored in the control under Axis MD parameter 32450[0]. To find the parameter, press “Setup” , make sure you are on page 1 according to the far-right corner of HSK, press “Mach data” then “Axis MD” . From the right VSK press “Search” and enter 32450 into the field. The backlash values may need to be adjusted over time to account for wear in the drive train system, see figure 4.4.3a. Figure 4.4.3a Axis MD Using the right VSK press “Select axis” and you will see the axis list appear. You need “Manufacturer” level password to be able to edit the fields, see figure 4.4.3b. Figure 4.4.3b Select Axis
原文摘录	5.4.5 Backlash Constants Backlash settings for the TCsi machines are set by the factory during the build for the X & Z-Axis. These values are unique for each machine. The backlash values for the two axes, AX1 (X-Axis) and AX2 (Z-Axis) are stored in the control under Axis MD parameter 32450[0]. To find the parameter, press “Setup” , make sure you are on page 1 according to the far-right corner of HSK, press “Mach data” then “Axis MD” . From the right VSK press “Search” and enter 32450 into the field. The backlash values may need to be adjusted over time to account for wear in the drive train system, see figure 4.4.3a. Figure 4.4.3a Axis MD Using the right VSK press “Select axis” and you will see the axis list appear. You need “Manufacturer” level password to be able to edit the fields, see figure 4.4.3b. Figure 4.4.3b Select Axis

#29 lty-sop-oil-skimmer			SOP 作业程序
SOP编号	SOP-TC820LTYsi-326	程序名称	泵式油水分离器排油
部件	pump type oil skimmer	章节	TC820LTYsi Model SOP
起始页	130	安全等级	high
知识类型	sop	适用机型	TC820LTYsi

摘要 | TC820LTYsi 泵式油水分离器排油：按手册页码执行并以验收条件确认结果。

A • 标识与来源

记录ID	lty-sop-oil-skimmer
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	TC820LTYsi
型号范围	TC820LTYsi
型号专属	是
章节编号	LTY-SOP

章节	TC820LTYsi Model SOP
起始页	130
结束页	131

C • 分类与治理

记录类别	sop
知识类型	sop
部件	pump type oil skimmer
安全等级	high
可入库	是
需人工复核	是

D • 业务明细

前置条件	· 已安装泵式撇油器 · 准备废油容器
安全警告	· 防止分离油溢回冷却液箱 · 拆泵/皮带前先急停断电并装防护
作业步骤	· 检查分离油液位。 · 打开阀门排入收集容器。 · 关闭阀门并按环保规定处置。 · 检查回流路径和溢漏。
验证方式	分离油未溢回冷却液箱，阀门和回流正常。
验收标准	分离油未溢回冷却液箱，阀门和回流正常。
SOP编号	SOP-TC820LTYsi-326
程序名称	泵式油水分离器排油
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	TC820LTYsi 泵式油水分离器排油：按手册页码执行并以验收条件确认结果。
关键词	· 撇油器 · 分离器 · 排油

检索文本

TC820LTYsi; sop; sop; pump type oil skimmer; 泵式油水分离器排油; TC820LTYsi 泵式油水分离器排油: 按手册页码执行并以验收条件确认结果。; 检查分离油液位。; 打开阀门排入收集容器。; 关闭阀门并按环保规定处置。; 检查回流路径和溢漏。; 撇油器; 分离器; 排油; 9.1.2 Pump Type Oil Skimmer The pump type oil skimmer on the TC820LTYsi machines is positioned adjacent to the coolant pump. The oil skimmer is wired directly to the electrical cabinet and is controlled using a button on the MCP or through the use of M-Codes. As the pump of the oil skimmer runs the top layer of the coolant is removed with the layer of oil and dumped in the oil separator. The oil and coolant collected by the oil skimmer is then separated with the coolant being returned back into the coolant tank and the oil passed into a section of the oil separator where it can be removed. The oil skimmer helps keep your coolant free of oil, extending the life of the coolant and reducing coolant maintenance. Figure 8.1.2 shows an overview of the pump ty pe oil skimmer on the TC820LTYsi machine. Figure 8.1.2 Pump Type Oil Skimmer 9.1.3 Maintenance 9.1.3.1 Emptying the Skimmed Oil Both oil skimmer options on the TCsi machines come with a collection container. • On the TC820si machines the skimmed oil will run off the oil separator into the collection container. There is a bucket beneath it to catch any excess oil, but it is important to empty the skimmed oil before it spills over the top. Simply remove the funnel and pick up the container using the handle and empty the oil where it can be disposed. Any oil skimmed while the tank is removed will be captured in the bucket below the tank. • On the TC820LTY machines the skimmed oil will be held in the oil separator. There is a valve that can be opened to empty the skimmed oil from the oil separator into the collection container. It is important to empty the skimmed oil from the oil separator into the collection container before it runs over the top where it will then return to the coolant tank. Simply open the valve to empty

原文摘录

9.1.2 Pump Type Oil Skimmer The pump type oil skimmer on the TC820LTYsi machines is positioned adjacent to the coolant pump. The oil skimmer is wired directly to the electrical cabinet and is controlled using a button on the MCP or through the use of M-Codes. As the pump of the oil skimmer runs the top layer of the coolant is removed with the layer of oil and dumped in the oil separator. The oil and coolant collected by the oil skimmer is then separated with the coolant being returned back into the coolant tank and the oil passed into a section of the oil separator where it can be removed. The oil skimmer helps keep your coolant free of oil, extending the life of the coolant and reducing coolant maintenance. Figure 8.1.2 shows an overview of the pump ty pe oil skimmer on the TC820LTYsi machine. Figure 8.1.2 Pump Type Oil Skimmer 9.1.3 Maintenance 9.1.3.1 Emptying the Skimmed Oil Both oil skimmer options on the TCsi machines come with a collection container. • On the TC820si machines the skimmed oil will run off the oil separator into the collection container. There is a bucket beneath it to catch any excess oil, but it is important to empty the skimmed oil before it spills over the top. Simply remove the funnel and pick up the container using the handle and empty the oil where it can be disposed. Any oil skimmed while the tank is removed will be captured in the bucket below the tank. • On the TC820LTY machines the skimmed oil will be held in the oil separator. There is a valve that can be opened to empty the skimmed oil from the oil separator into the collection container. It is important to empty the skimmed oil from the oil separator into the collection container before it runs over the top where it will then return to the coolant tank. Simply open the valve to empty

#30 lty-sop-part-catcher			SOP 作业程序
SOP编号	SOP-TC820LTYsi-327	程序名称	接料器清洁和功能检查
部件	part catcher	章节	TC820LTYsi Model SOP
起始页	132	安全等级	high
知识类型	sop	适用机型	TC820LTYsi

摘要 | TC820LTYsi 接料器清洁和功能检查: 按手册页码执行并以验收条件确认结果。

A • 标识与来源

记录ID	lty-sop-part-catcher
文档ID	trak-tcsi-service-34242-v022725

文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	TC820LTYsi
型号范围	TC820LTYsi
型号专属	是
章节编号	LTY-SOP
章节	TC820LTYsi Model SOP
起始页	132
结束页	133

C • 分类与治理

记录类别	sop
知识类型	sop
部件	part catcher
安全等级	high
可入库	是
需人工复核	是

D • 业务明细

前置条件	生产结束且接料器处于安全状态
安全警告	· 停止动作后清洁 · 堵塞回流路径会导致冷却液溢出
作业步骤	· 清除接料器本体和滑道切屑。 · 清空储料桶，防止拥堵。 · 清除桶内和筛网下方切屑。 · 用 MCP 测试升起和闲置位置及反馈。
验证方式	升降无迟滞、位置反馈正确、回流畅通。
验收标准	升降无迟滞、位置反馈正确、回流畅通。
SOP编号	SOP-TC820LTYsi-327
程序名称	接料器清洁和功能检查
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	TC820LTYsi 接料器清洁和功能检查：按手册页码执行并以验收条件确认结果。
------	--

关键词	· 接料器 · 切屑 · 冷却液回流
检索文本	TC820LTYsi; sop; sop; part catcher; 接料器清洁和功能检查; TC820LTYsi 接料器清洁和功能检查: 按手册页码执行并以验收条件确认结果。; 清除接料器本体和滑道切屑。; 清空储料桶, 防止拥堵。; 清除桶内和筛网下方切屑。; 用 MCP 测试升起和闲置位置及反馈。; 接料器; 切屑; 冷却液回流; 5. Loosen or remove as needed the two screws holding the pulley bracket see figure 8.1.2.2c in order to compress the spring and remove the belt. Figure 8.1.2.2a Mounting Bolts Figure 8.1.2.2b Cover Screws Figure 8.1.2.2c Pulley 5. Replace the belt and reverse the above procedure to replace the oil skimmer. 9.2 Part Catcher 9.2.1 Programable Part Catcher The TRAK part catcher on the TCsi machines is operated using the button on the MCP or through the use M-Codes embedded in a program. Driven by the hydraulic system, the part catcher can be deployed to catch finished parts as they are cut off and retracted to be clear during machining operations. Utilizing the part catcher can aid with unattended machining as the finished part runs down the shoot to be stored in a built-in bucket, see figure 8.2.1.
原文摘录	5. Loosen or remove as needed the two screws holding the pulley bracket see figure 8.1.2.2c in order to compress the spring and remove the belt. Figure 8.1.2.2a Mounting Bolts Figure 8.1.2.2b Cover Screws Figure 8.1.2.2c Pulley 5. Replace the belt and reverse the above procedure to replace the oil skimmer. 9.2 Part Catcher 9.2.1 Programable Part Catcher The TRAK part catcher on the TCsi machines is operated using the button on the MCP or through the use M-Codes embedded in a program. Driven by the hydraulic system, the part catcher can be deployed to catch finished parts as they are cut off and retracted to be clear during machining operations. Utilizing the part catcher can aid with unattended machining as the finished part runs down the shoot to be stored in a built-in bucket, see figure 8.2.1.

#31 lty-sop-hp-pump			SOP 作业程序
SOP编号	SOP-TC820LTYsi-328	程序名称	高压冷却泵更换
部件	high-pressure coolant	章节	TC820LTYsi Model SOP
起始页	134	安全等级	critical
知识类型	sop	适用机型	TC820LTYsi

摘要 | TC820LTYsi 高压冷却泵更换: 按手册页码执行并以验收条件确认结果。

A • 标识与来源

记录ID	lty-sop-hp-pump
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	TC820LTYsi
型号范围	TC820LTYsi

型号专属	是
章节编号	LTY-SOP
章节	TC820LTYsi Model SOP
起始页	134
结束页	135

C • 分类与治理

记录类别	sop
知识类型	sop
部件	high-pressure coolant
安全等级	critical
可入库	是
需人工复核	是

D • 业务明细

前置条件	· 确认高压泵故障 · 急停/断电并准备接液和起吊设备
安全警告	· 约 220 psi，先关闭流体和电源并排液 · 泵总成是重物必须起吊
作业步骤	· MCP 关闭流体并关闭电源。 · 排液至过滤器罐清空。 · 断开泵电缆和传感器线并记录方向。 · 拆过滤器、螺钉并吊出旧泵。 · 装入新泵，复接管线和电缆。 · 启动后核对旋向，至少无故障运行 10 秒并查漏。
验证方式	约 220 psi、旋向正确、10 秒无故障且无泄漏。
验收标准	约 220 psi、旋向正确、10 秒无故障且无泄漏。
SOP编号	SOP-TC820LTYsi-328
程序名称	高压冷却泵更换
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	TC820LTYsi 高压冷却泵更换：按手册页码执行并以验收条件确认结果。
关键词	· 220 psi · 起吊 · 10 秒

检索文本	TC820LTYsi; sop; sop; high-pressure coolant; 高压冷却泵更换; TC820LTYsi 高压冷却泵更换：按手册页码执行并以验收条件确认结果。; MCP 关闭流体并关闭电源。; 排液至过滤器罐清空。; 断开泵电缆和传感器线并记录方向。; 拆过滤器、螺钉并吊出旧泵。; 装入新泵，复接管线和电缆。; 启动后核对旋向，至少无故障运行 10 秒并查漏。; 220 psi; 起吊; 10 秒; 9.3.1 High Pressure Coolant The TRAK high pressure coolant system, see figure 8.3.1, delivers coolant at 220 psi directly through the turret and cutting tool effectively dissipating chips and heat during machining. The increased pressure will allow for heavier cuts, higher feed rates, deep-hole drilling, extended tool life and better surface finishes. This produces reduced cycle times, increase throughput, and higher-quality parts. The high-pressure coolant pump is fitted with a filter and pressure sensor, this helps keep the filter clean and at enough pressure to ensure optimal cutting conditions. Figure 8.3.1 High Pressure Coolant 9.3.2 High Pressure Coolant Pump The high-pressure coolant system uses a larger pump delivering coolant at 220 psi. If the pump fails it will need to be replaced. Warning!!! Heavy object to be installed, proper safety precautions are needed. To replace a faulty pump: 1. Turn off fluid flow using button on the MCP and turn off power to the TCsi machine. 2. Drain enough coolant to clear the filter canister. 3. Disconnect the cable going from the electric cabinet to coolant pump and the wires at the pressure sensor, see figure 8.3.2. Note orientation of the sensor wires.
原文摘录	9.3.1 High Pressure Coolant The TRAK high pressure coolant system, see figure 8.3.1, delivers coolant at 220 psi directly through the turret and cutting tool effectively dissipating chips and heat during machining. The increased pressure will allow for heavier cuts, higher feed rates, deep-hole drilling, extended tool life and better surface finishes. This produces reduced cycle times, increase throughput, and higher-quality parts. The high-pressure coolant pump is fitted with a filter and pressure sensor, this helps keep the filter clean and at enough pressure to ensure optimal cutting conditions. Figure 8.3.1 High Pressure Coolant 9.3.2 High Pressure Coolant Pump The high-pressure coolant system uses a larger pump delivering coolant at 220 psi. If the pump fails it will need to be replaced. Warning!!! Heavy object to be installed, proper safety precautions are needed. To replace a faulty pump: 1. Turn off fluid flow using button on the MCP and turn off power to the TCsi machine. 2. Drain enough coolant to clear the filter canister. 3. Disconnect the cable going from the electric cabinet to coolant pump and the wires at the pressure sensor, see figure 8.3.2. Note orientation of the sensor wires.

#32 lty-sop-hp-sensor		SOP 作业程序	
SOP编号	SOP-TC820LTYsi-329	程序名称	高压冷却压力传感器更换
部件	high-pressure coolant	章节	TC820LTYsi Model SOP
起始页	135	安全等级	critical
知识类型	sop	适用机型	TC820LTYsi

摘要 | TC820LTYsi 高压冷却压力传感器更换：按手册页码执行并以验收条件确认结果。

A • 标识与来源

记录ID	lty-sop-hp-sensor
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	TC820LTYsi
型号范围	TC820LTYsi
型号专属	是
章节编号	LTY-SOP
章节	TC820LTYsi Model SOP
起始页	135
结束页	135

C • 分类与治理

记录类别	sop
知识类型	sop
部件	high-pressure coolant
安全等级	critical
可入库	是
需人工复核	是

D • 业务明细

前置条件	· 流体已关闭且电源隔离 · 记录原线方向
安全警告	· 未释放压力不得拆卸 · 更换后查漏
作业步骤	· 关闭流体和电源。 · 移开滑盖并拆 captive screw。 · 断开两个 ferrule，记录方向并拆传感器。 · 安装新传感器。 · 设为 85 psi，启动冷却至少 10 秒查漏。
验证方式	85 psi、反馈正常且无泄漏。
验收标准	85 psi、反馈正常且无泄漏。
SOP编号	SOP-TC820LTYsi-329
程序名称	高压冷却压力传感器更换
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	TC820LTYsi 高压冷却压力传感器更换：按手册页码执行并以验收条件确认结果。
关键词	· 85 psi · 压力传感器 · 高压冷却

检索文本

TC820LTYsi; sop; sop; high-pressure coolant; 高压冷却压力传感器更换; TC820LTYsi 高压冷却压力传感器更换: 按手册页码执行并以验收条件确认结果。; 关闭流体和电源。; 移开滑盖并拆 captive screw。; 断开两个 ferrule, 记录方向并拆传感器。; 安装新传感器。; 设为 85 psi, 启动冷却至少 10 秒查漏。; 85 psi; 压力传感器; 高压冷却; Figure 8.3.2 High Pressure Coolant Wiring 4. Prepare to collect the coolant and then disconnect the coolant hose from the pressure sensor. 5. Remove the filter with pressure sensor from the coolant pump. 6. Remove four SHCS screws that bolt the motor to the sheet metal base. 7. Using a proper lifting device, slowly lift the pump assembly from the coolant tank allowing it to drain thoroughly. Set down in a safe manner. 8. Using the lifting device, carefully guide the new pump assembly into position and reconnect the four SHCS. 9. Reconnect the filter and pressure sensor assembly to the pump outlet. The sensor should not need recalibrating. 10. Reconnect coolant hose using Teflon tape. 11. Connect wires to the pressure sensor. Plug in the pump electric cable to electric cabinet. 12. Power up the TCsi machine and turn on the coolant using the button on the MCP. Verify the motor is turning in the correct direction with the arrows on top of the pump and make sure the coolant comes on and operates for at least 10 seconds without faulting. 9.3.3 High Pressure Coolant Pressure Sensor The coolant pressure sensor checks for proper coolant pressure through the system. The coolant pressure sensor is designed to warn if there is insufficient pressure in the system, it monitors the coolant pressure in case of coolant runout or equipment failure. Under these conditions there is a warning message displayed to give the operator a chance to correct the problem. If this warning message is triggered, the sensor halts the current program and stops operation in the machine. 9.3.3.1 Pressure sensor replacement: 1. Turn off fluid flow using the button on the MCP and turn off power to the TCsi machine. 2. Loosen the slide cap and unscrew the captive screw at the top of the sensor. Now you can slide the cap out of the way, see figure 8.3.2. 3. Disconnect two small electric ferrules on the sensor and remove the wires then loosen the sensor with a wrench. Note the wire orientation. Replace the old sensor with the new one. 4. Reverse above steps to reconnect the sensor. 5. Set the pressure value of the pressure sensor to 85psi (6 kg/cm2). 6. Power up the TCsi machine and turn on the coolant using the button on the MCP. Make sure the coolant comes on and operates for at least 10 seconds without faulting. 9.3.4 High Pressure Coolant Filter Element The filter element is an important safeguard to keep contaminants out of the coolant system. 9.3.4.1 Filter Element Replacement: The filter for the high-pressure coolant system should be cleaned every time the pressure sensor triggers or you replace your coolant:

原文摘录

Figure 8.3.2 High Pressure Coolant Wiring 4. Prepare to collect the coolant and then disconnect the coolant hose from the pressure sensor. 5. Remove the filter with pressure sensor from the coolant pump. 6. Remove four SHCS screws that bolt the motor to the sheet metal base. 7. Using a proper lifting device, slowly lift the pump assembly from the coolant tank allowing it to drain thoroughly. Set down in a safe manner. 8. Using the lifting device, carefully guide the new pump assembly into position and reconnect the four SHCS. 9. Reconnect the filter and pressure sensor assembly to the pump outlet. The sensor should not need recalibrating. 10. Reconnect coolant hose using Teflon tape. 11. Connect wires to the pressure sensor. Plug in the pump electric cable to electric cabinet. 12. Power up the TCsi machine and turn on the coolant using the button on the MCP. Verify the motor is turning in the correct direction with the arrows on top of the pump and make sure the coolant comes on and operates for at least 10 seconds without faulting. 9.3.3 High Pressure Coolant Pressure Sensor The coolant pressure sensor checks for proper coolant pressure through the system. The coolant pressure sensor is designed to warn if there is insufficient pressure in the system, it monitors the coolant pressure in case of coolant runout or equipment failure. Under these conditions there is a warning message displayed to give the operator a chance to correct the problem. If this warning message is triggered, the sensor halts the current program and stops operation in the machine. 9.3.3.1 Pressure sensor replacement: 1. Turn off fluid flow using the button on the MCP and turn off power to the TCsi machine. 2. Loosen the slide cap and unscrew the captive screw at the top of the sensor. Now you can slide the cap out of the way, see figure 8.3.2. 3. Disconnect two small electric ferrules on the sensor and remove the wires then loosen the sensor with a wrench. Note the wire orientation. Replace the old sensor with the new one. 4. Reverse above steps to reconnect the sensor. 5. Set the pressure value of the pressure sensor to 85psi (6 kg/cm2). 6. Power up the TCsi machine and turn on the coolant using the button on the MCP. Make sure the coolant comes on and operates for at least 10 seconds without faulting. 9.3.4 High Pressure Coolant Filter Element The filter element is an important safeguard to keep contaminants out of the coolant system. 9.3.4.1 Filter Element Replacement: The filter for the high-pressure coolant system should be cleaned every time the pressure sensor triggers or you replace your coolant:

#33 lty-sop-hp-filter			SOP 作业程序
SOP编号	SOP-TC820LTYsi-330	程序名称	高压冷却过滤元件更换
部件	high-pressure coolant	章节	TC820LTYsi Model SOP
起始页	135	安全等级	high
知识类型	sop	适用机型	TC820LTYsi

摘要 | TC820LTYsi 高压冷却过滤元件更换：按手册页码执行并以验收条件确认结果。

A • 标识与来源

记录ID	lty-sop-hp-filter
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
覆盖型号	TC820LTYsi
型号范围	TC820LTYsi
型号专属	是
章节编号	LTY-SOP
章节	TC820LTYsi Model SOP
起始页	135
结束页	136

C • 分类与治理

记录类别	sop
知识类型	sop
部件	high-pressure coolant
安全等级	high
可入库	是
需人工复核	是

D • 业务明细

前置条件	· 压力传感器触发或更换冷却液后执行 · 系统断电隔离
安全警告	· 保护 O-ring 防止泄漏 · 通电前确认滤罐和压力表牢固

作业步骤	· 关闭系统并拆压力传感器。 · 拆滤器顶部四个螺钉并托住壳体。 · 取出滤芯，检查并润滑 O-ring。 · 装新滤芯、滤罐并交叉紧固。 · 用 Teflon tape 复装传感器，通电查漏。
验证方式	O-ring、滤芯、密封和泄漏检查通过。
验收标准	O-ring、滤芯、密封和泄漏检查通过。
SOP编号	SOP-TC820LTYsi-330
程序名称	高压冷却过滤元件更换
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	TC820LTYsi 高压冷却过滤元件更换：按手册页码执行并以验收条件确认结果。
关键词	· O-ring · 滤芯 · Teflon tape
检索文本	TC820LTYsi; sop; sop; high-pressure coolant; 高压冷却过滤元件更换; TC820LTYsi 高压冷却过滤元件更换：按手册页码执行并以验收条件确认结果。; 关闭系统并拆压力传感器。; 拆滤器顶部四个螺钉并托住壳体。; 取出滤芯，检查并润滑 O-ring。; 装新滤芯、滤罐并交叉紧固。; 用 Teflon tape 复装传感器，通电查漏。; O-ring; 滤芯; Teflon tape; Figure 8.3.2 High Pressure Coolant Wiring 4. Prepare to collect the coolant and then disconnect the coolant hose from the pressure sensor. 5. Remove the filter with pressure sensor from the coolant pump. 6. Remove four SHCS screws that bolt the motor to the sheet metal base. 7. Using a proper lifting device, slowly lift the pump assembly from the coolant tank allowing it to drain thoroughly. Set down in a safe manner. 8. Using the lifting device, carefully guide the new pump assembly into position and reconnect the four SHCS. 9. Reconnect the filter and pressure sensor assembly to the pump outlet. The sensor should not need recalibrating. 10. Reconnect coolant hose using Teflon tape. 11. Connect wires to the pressure sensor. Plug in the pump electric cable to electric cabinet. 12. Power up the TCsi machine and turn on the coolant using the button on the MCP. Verify the motor is turning in the correct direction with the arrows on top of the pump and make sure the coolant comes on and operates for at least 10 seconds without faulting. 9.3.3 High Pressure Coolant Pressure Sensor The coolant pressure sensor checks for proper coolant pressure through the system. The coolant pressure sensor is designed to warn if there is insufficient pressure in the system, it monitors the coolant pressure in case of coolant runout or equipment failure. Under these conditions there is a warning message displayed to give the operator a chance to correct the problem. If this warning message is triggered, the sensor halts the current program and stops operation in the machine. 9.3.3.1 Pressure sensor replacement: 1. Turn off fluid flow using the button on the MCP and turn off power to the TCsi machine. 2. Loosen the slide cap and unscrew the captive screw at the top of the sensor. Now you can slide the cap out of the way, see figure 8.3.2. 3. Disconnect two small electric ferrules on the sensor and remove the wires then loosen the sensor with a wrench. Note the wire orientation. Replace the old sensor with the new one. 4. Reverse above steps to reconnect the sensor. 5. Set the pressure value of the pressure sensor to 85psi (6 kg/cm2). 6. Power up the TCsi machine and turn on the coolant using the button on the MCP. Make sure the coolant comes on and operates for at least 10 seconds without faulting. 9.3.4 High Pressure Coolant Filter Element The filter element is an important safeguard to keep contaminants out of the coolant system. 9.3.4.1 Filter Element Replacement: The filter for the high-pressure coolant system should be cleaned every time the pressure sensor triggers or you replace your coolant:

原文摘录

Figure 8.3.2 High Pressure Coolant Wiring 4. Prepare to collect the coolant and then disconnect the coolant hose from the pressure sensor. 5. Remove the filter with pressure sensor from the coolant pump. 6. Remove four SHCS screws that bolt the motor to the sheet metal base. 7. Using a proper lifting device, slowly lift the pump assembly from the coolant tank allowing it to drain thoroughly. Set down in a safe manner. 8. Using the lifting device, carefully guide the new pump assembly into position and reconnect the four SHCS. 9. Reconnect the filter and pressure sensor assembly to the pump outlet. The sensor should not need recalibrating. 10. Reconnect coolant hose using Teflon tape. 11. Connect wires to the pressure sensor. Plug in the pump electric cable to electric cabinet. 12. Power up the TCsi machine and turn on the coolant using the button on the MCP. Verify the motor is turning in the correct direction with the arrows on top of the pump and make sure the coolant comes on and operates for at least 10 seconds without faulting.

9.3.3 High Pressure Coolant Pressure Sensor The coolant pressure sensor checks for proper coolant pressure through the system. The coolant pressure sensor is designed to warn if there is insufficient pressure in the system, it monitors the coolant pressure in case of coolant runout or equipment failure. Under these conditions there is a warning message displayed to give the operator a chance to correct the problem. If this warning message is triggered, the sensor halts the current program and stops operation in the machine.

9.3.3.1 Pressure sensor replacement: 1. Turn off fluid flow using the button on the MCP and turn off power to the TCsi machine. 2. Loosen the slide cap and unscrew the captive screw at the top of the sensor. Now you can slide the cap out of the way, see figure 8.3.2. 3. Disconnect two small electric ferrules on the sensor and remove the wires then loosen the sensor with a wrench. Note the wire orientation. Replace the old sensor with the new one. 4. Reverse above steps to reconnect the sensor. 5. Set the pressure value of the pressure sensor to 85psi (6 kg/cm2). 6. Power up the TCsi machine and turn on the coolant using the button on the MCP. Make sure the coolant comes on and operates for at least 10 seconds without faulting.

9.3.4 High Pressure Coolant Filter Element The filter element is an important safeguard to keep contaminants out of the coolant system.

9.3.4.1 Filter Element Replacement: The filter for the high-pressure coolant system should be cleaned every time the pressure sensor triggers or you replace your coolant:

#34 lty-sop-lns			SOP 作业程序
SOP编号	SOP-TC820LTYsi-331	程序名称	LNS 棒料送料器安全操作和故障检查
部件	barfeeder	章节	TC820LTYsi Model SOP
起始页	136	安全等级	critical
知识类型	sop	适用机型	TC820LTYsi

摘要 | TC820LTYsi LNS 棒料送料器安全操作和故障检查：按手册页码执行并以验收条件确认结果。

A • 标识与来源

记录ID	lty-sop-lns
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725
语言	en
产品系列	TRAK TCsi

B • 型号与位置

设备型号	TC820LTYsi
------	------------

覆盖型号	TC820LTYsi
型号范围	TC820LTYsi
型号专属	是
章节编号	LTY-SOP
章节	TC820LTYsi Model SOP
起始页	136
结束页	140

C • 分类与治理

记录类别	sop
知识类型	sop
部件	barfeeder
安全等级	critical
可入库	是
需人工复核	是

D • 业务明细

前置条件	· 已安装 Barfeed Ready · 操作员完成 LNS/机床安全培训
安全警告	· 维修仅由 TRAK/LNS 受训人员执行 · 禁止手伸入 magazine、保护盖和送料机构 · 急停和磁性安全开关必须有效
作业步骤	· 检查主电源、急停、停止灯和磁性安全开关。 · 检查主入口盖、保护盖和锁扣。 · 核对棒料 6–80 mm、300–1600 mm。 · 排查 I6.0/I6.2/I6.6 和 Easy Extend 配置。 · 无法恢复时升级受训维修。
验证方式	安全开关、反馈、选项配置和试运行均正常。
验收标准	安全开关、反馈、选项配置和试运行均正常。
SOP编号	SOP-TC820LTYsi-331
程序名称	LNS 棒料送料器安全操作和故障检查
适用机型	TC820LTYsi

E • 检索与原文

中文摘要	TC820LTYsi LNS 棒料送料器安全操作和故障检查：按手册页码执行并以验收条件确认结果。
关键词	· LNS · 6-80 mm · 300-1600 mm · I6.0 · I6.2 · I6.6

检索文本	TC820LTYsi; sop; sop; barfeeder; LNS 棒料送料器安全操作和故障检查; TC820LTYsi LNS 棒料送料器安全操作和故障检查: 按手册页码执行并以验收条件确认结果。; 检查主电源、急停、停止灯和磁性安全开关。; 检查主入口盖、保护盖和锁扣。; 核对棒料 6–80 mm、300–1600 mm。; 排查 I6.0/I6.2/I6.6 和 Easy Extend 配置。; 无法恢复时升级受训维修。; LNS; 6-80 mm; 300-1600 mm; I6.0; I6.2; I6.6; 1. Turn off system power. 2. Remove pressure sensor, see section 8.3.3.1 Carefully remove gauge with a wrench, see figure 8.3.4.1a. 3. Remove the four socket head cap screws from the top of the filter body, be ready to catch the element housing and don’ t lose the O- ring. 4. Slide off the filter element from the housing. 5. Check the canister O-ring and replace if needed. Figure 8.3.4.1a Gauge 6. Apply lubricant to the O-ring with petroleum jelly or light oil and place on filter head. 7. Slide the new filter on the housing, install canister and make sure O- ring is in its groove, see figure 8.3.4.1b. 8. Cross tighten the screws. 9. Replace the gauge and pressure sensor using Teflon tape. 10. Confirm there are no leaks after powering the system. Figure 8.3.4.1b O-ring 9.4 LNS Quick Load Servo 80 S2 Barfeed and Barfeed Ready 9.4.1 LNS Barfeed Operation The LNS Quick Load Servo 80 S2 Bar Feeder allows for continuous machining. With quick changeovers from one part run to another and a library of up to 500-part programs, this servo driven bar feed will drastically increase productivity. Equipped with an automatic diameter changing system, no manual adjustment is required reducing changeover time to a minimum. The LNS servo motor control technology guarantees accurate bar feeding into the machine and prevents hitting the turret with its soft touch positioning.
原文摘录	1. Turn off system power. 2. Remove pressure sensor, see section 8.3.3.1 Carefully remove gauge with a wrench, see figure 8.3.4.1a. 3. Remove the four socket head cap screws from the top of the filter body, be ready to catch the element housing and don’ t lose the O- ring. 4. Slide off the filter element from the housing. 5. Check the canister O-ring and replace if needed. Figure 8.3.4.1a Gauge 6. Apply lubricant to the O-ring with petroleum jelly or light oil and place on filter head. 7. Slide the new filter on the housing, install canister and make sure O- ring is in its groove, see figure 8.3.4.1b. 8. Cross tighten the screws. 9. Replace the gauge and pressure sensor using Teflon tape. 10. Confirm there are no leaks after powering the system. Figure 8.3.4.1b O-ring 9.4 LNS Quick Load Servo 80 S2 Barfeed and Barfeed Ready 9.4.1 LNS Barfeed Operation The LNS Quick Load Servo 80 S2 Bar Feeder allows for continuous machining. With quick changeovers from one part run to another and a library of up to 500-part programs, this servo driven bar feed will drastically increase productivity. Equipped with an automatic diameter changing system, no manual adjustment is required reducing changeover time to a minimum. The LNS servo motor control technology guarantees accurate bar feeding into the machine and prevents hitting the turret with its soft touch positioning.

#35 lty-sop-chuck		SOP 作业程序	
SOP编号	SOP-TC820LTYsi-332	程序名称	三爪液压卡盘拆装
部件	hydraulic chuck	章节	TC820LTYsi Model SOP
起始页	66	安全等级	critical
知识类型	sop	适用机型	TC820LTYsi

摘要 | TC820LTYsi 三爪液压卡盘拆装: 按手册页码执行并以验收条件确认结果。

A • 标识与来源	
记录ID	lty-sop-chuck
文档ID	trak-tcsi-service-34242-v022725
文档标题	TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List
来源类型	maintenance_manual
源文件	维修手册/维修手册.pdf
制造商	Southwestern Industries, Inc.
文档版本	022725

语言	en
产品系列	TRAK TCsi
B • 型号与位置	
设备型号	TC820LTYsi
覆盖型号	TC820LTYsi
型号范围	TC820LTYsi
型号专属	是
章节编号	LTY-SOP
章节	TC820LTYsi Model SOP
起始页	66
结束页	66
C • 分类与治理	
记录类别	sop
知识类型	sop
部件	hydraulic chuck
安全等级	critical
可入库	是
需人工复核	是
D • 业务明细	
前置条件	· 卡盘需要拆装且已急停 · 准备吊装设备和吊环
安全警告	· 卡盘是重物必须吊装 · 安装后六螺钉 53.5 ft-lbs
作业步骤	· 松开卡盘并急停，拆除卡爪。 · 用吊环起吊卡盘，拆下 draw nut。 · 清洁安装面、检查高点并涂防咬合剂。 · 用吊装设备对齐 drawtube，手紧六个螺钉。 · 按 53.5 ft-lbs 紧固并完成复装。
验证方式	安装面、扭矩、夹紧和运行测试通过。
验收标准	安装面、扭矩、夹紧和运行测试通过。
SOP编号	SOP-TC820LTYsi-332
程序名称	三爪液压卡盘拆装
适用机型	TC820LTYsi
E • 检索与原文	
中文摘要	TC820LTYsi 三爪液压卡盘拆装：按手册页码执行并以验收条件确认结果。
关键词	· 卡盘 · 53.5 ft-lbs · drawtube

检索文本

TC820LTYsi; sop; sop; hydraulic chuck; 三爪液压卡盘拆装; TC820LTYsi 三爪液压卡盘拆装: 按手册页码执行并以验收条件确认结果。; 松开卡盘并急停, 拆除卡爪。; 用吊环起吊卡盘, 拆下 draw nut。; 清洁安装面、检查高点并涂防咬合剂。; 用吊装设备对齐 drawtube, 手紧六个螺钉。; 按 53.5 ft-lbs 紧固并完成复装。; 卡盘; 53.5 ft-lbs; drawtube; Installation Checklist • Press the oil skim button to make sure the skimmer starts to skim oil. • Verify the oil skimmer pump is rotating in the clockwise direction. • Press the oil skim button again to make sure the skimmer stops. • Attach the way lube runoff hose to the oil separator box. Part Catcher – See part catcher section 9.2 of the Service Manual. 38 • Test functionality using buttons on MCP. • • Press the parts button to make sure the part catcher moves to its raised position, press button again to make sure it returns to its idle position. High Pressure Coolant – See high pressure coolant section 8.3. • Wire up the high-pressure coolant pressure sensor. • Test filter pressure sensor: o With the pump turned on and the pressure sensor unplugged check the IO screen in TRAK diagnostic to see coolant pressure switch is false. 39 • Calibrate the coolant pressure sensor: o Remove any tool from the turret prior to calibration. o Using a screwdriver slowly tighten the screw on the top center of the sensor approximately 1/8th of a turn per second. o When the machine faults out, loosen the screw ¼ turn, and clear the fault on the pendent. Run the high-pressure coolant pump for 10 minutes and check for leaks. 3.15 Jaw Hydraulic Chuck Mounting 3.15.1 Instructions to remove the 3 Jaw Chuck: 1. Unclamp the chuck and press the E-Stop in center of travel. 2. Remove 2 of the 3 jaws from the chuck. 3. Using the remaining jaw to hold the chuck in place loosen all 6 screws holding the chuck to the spindle face. 4. Remove the last jaw from the chuck. 5. Remove chip cover from the center of the chuck. 6. Attach lifting eye bolt and hoist the chuck to take pressure off the drawtube and nut. 7. Remove the 6 screws holding the chuck. 8. Unscrew draw nut by hand, you may need to adjust the hoist until the nut rotates by hand. 3.15.2 Instructions to reinstall the 3 Jaw Chuck: 1. Clean off all the mounting surfaces and threads with WD40 and make sure they are free from any chips or swarf. 2. Use a fine grit deburring stone to check for dinks or high spots on the mounting faces. 3. Apply anti seize to the threads and mounting screws. 4. Set the draw tube to the mid travel position and E-stop the machine. 5. Using the eyebolt on the chuck, hoist it into position and align with the draw nut with the drawtube. 6. Screw the draw nut by hand until it is approximately ¾ engaged on the drawtube thread. 7. Align the chuck body holes to the holes on the spindle face and hand tighten the 6 screws. 8. Remove the hoist so the chuck is supporting itself. 9. Attach one jaw to the chuck and use it to torque the 6 mounting screws to 53.5 ft lbs. 10. Remove the jaw again from the chuck and thread the draw nut until it bottoms against the bottom of the drawtube thread.

原文摘录

Installation Checklist • Press the oil skim button to make sure the skimmer starts to skim oil. • Verify the oil skimmer pump is rotating in the clockwise direction. • Press the oil skim button again to make sure the skimmer stops. • Attach the way lube runoff hose to the oil separator box. Part Catcher – See part catcher section 9.2 of the Service Manual. 38 • Test functionality using buttons on MCP. • • Press the parts button to make sure the part catcher moves to its raised position, press button again to make sure it returns to its idle position. High Pressure Coolant – See high pressure coolant section 8.3. • Wire up the high-pressure coolant pressure sensor. • Test filter pressure sensor: o With the pump turned on and the pressure sensor unplugged check the IO screen in TRAK diagnostic to see coolant pressure switch is false. 39 • Calibrate the coolant pressure sensor: • o Remove any tool from the turret prior to calibration. o Using a screwdriver slowly tighten the screw on the top center of the sensor approximately 1/8th of a turn per second. o When the machine faults out, loosen the screw ¼ turn, and clear the fault on the pendent. Run the high-pressure coolant pump for 10 minutes and check for leaks. 3.15 Jaw Hydraulic Chuck Mounting 3.15.1 Instructions to remove the 3 Jaw Chuck: 1. Unclamp the chuck and press the E-Stop in center of travel. 2. Remove 2 of the 3 jaws from the chuck. 3. Using the remaining jaw to hold the chuck in place loosen all 6 screws holding the chuck to the spindle face. 4. Remove the last jaw from the chuck. 5. Remove chip cover from the center of the chuck. 6. Attach lifting eye bolt and hoist the chuck to take pressure off the drawtube and nut. 7. Remove the 6 screws holding the chuck. 8. Unscrew draw nut by hand, you may need to adjust the hoist until the nut rotates by hand. 3.15.2 Instructions to reinstall the 3 Jaw Chuck: 1. Clean off all the mounting surfaces and threads with WD40 and make sure they are free from any chips or swarf. 2. Use a fine grit deburring stone to check for dinks or high spots on the mounting faces. 3. Apply anti seize to the threads and mounting screws. 4. Set the draw tube to the mid travel position and E-stop the machine. 5. Using the eyebolt on the chuck, hoist it into position and align with the draw nut with the drawtube. 6. Screw the draw nut by hand until it is approximately ¾ engaged on the drawtube thread. 7. Align the chuck body holes to the holes on the spindle face and hand tighten the 6 screws. 8. Remove the hoist so the chuck is supporting itself. 9. Attach one jaw to the chuck and use it to torque the 6 mounting screws to 53.5 ft lbs. 10. Remove the jaw again from the chuck and thread the draw nut until it bottoms against the bottom of the drawtube thread.